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Highlights

- An adaptive phase-field method is proposed based on trilinear multi-node elements.
- The length scale insensitive phase-field model is combined with the adaptive method.
- A robust adaptive criterion associated with the internal length scale is outlined.
- Substantial computational memory and time required in phase-field modelling are reduced.
- The method is successfully used to simulate simple and mixed-mode fracture in three-dimensional conditions.

An efficient adaptive length scale insensitive phase-field method for three-dimensional fracture of solids using trilinear multi-node elements

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Abstract

As a diffused fracture theory, phase-field models can seamlessly simulate complex crack patterns such as extending, branching, and merging. Despite the success of phase-field models, there are two issues in previous methods of three-dimensional (3-D) fracture. Firstly, the nonlinear governing equations lead to the huge computational costs, which hinder the application of phase-field models in 3-D problems. Secondly, these models, which are mostly developed based on a simple quadratic degradation function, provide numerical solutions that are sensitive to a length scale. Hence, this work addresses an efficient adaptive phase-field model with the aid of trilinear multi-node elements. The order of the elements remains constant with the increase of the number of nodes. As the mesh size and length scale significantly influence the numerical precision, a robust adaptive criterion is established in which the element refinement is controlled by both internal length scale and phase-field. According to the criterion, an expected mesh density in the failure domain can be obtained even for nonuniform initial mesh. Besides, being able to extend the phase-field regularized cohesive zone model, the adaptive model provides length scale insensitive responses for both crack path and peak load. The failure of brittle and quasi-brittle materials in three-dimensional conditions, including simple and mixed-mode fracture, can be simulated by the

proposed model. Several benchmark examples are analyzed to show the efficiency and accuracy of the trilinear element-based adaptive phase-field model (TAPFM), and the results are compared with the standard phase-field model as well as experimental data.

Keywords: Phase-field model; Fracture; Trilinear multi-node elements; Adaptive strategy; Three-dimensional modelling.

1. Introduction

Fracture is one of the most commonly encountered failure modes for solid materials [1], which often plays an important role in the origination or end of engineering accidents. Consequently, the prediction of fracture is one of the major concerns in structural designs in many fields, e.g., aerospace industry, geology, civil and hydraulic engineering. With the development of fracture mechanics, numerical modelling has become an indispensable method to reveal the mechanism of crack initiation and evolution.

During recent decades, various kinds of numerical models were developed to obtain the fracturing process, which can be generally classified into two categories, i.e., discontinuous and continuous approaches [2]. Discontinuous methods represent the crack's topology by applying tracking methods to introduce new boundaries, such as interface elements [3] or level sets [4]. Popular discontinuous models include the extended finite element method (XFEM) [5], discrete element method [6], boundary element method [7], cohesive zone models [8], and the phantom-node method [9]. To obtain an exact crack path, most of the methods require additional criteria to determine where the new boundaries should be generated. Besides, the strong discontinuity leads to a singularity problem on the crack tip, which usually demands complex solutions to obtain an accurate displacement field.

On the contrary, the continuous approaches do not demand the separated crack surface and discontinuous displacement field. These methods usually smear the fracture in a localization band with a finite width so that the displacement jump can be expressed through its gradient in the damaged zone. Among them, the representative continuous models are gradient damage models [10], peridynamic models [11], non-local integral damage models [12], and phase-field models [13-16]. Possessing the capacity of naturally obtaining the complex crack evolution from the governing equations, such as crack initiation, propagation, coalescence, and branching, phase-field model (PFM) became a rapidly developed theory in the past decade. In the model, the crack is

diffused using a continuum scalar field, i.e. phase-field [17], which can be regarded as an independent physical field to represent the extent of damage.

Due to the extremely fine mesh required to determine a proper smeared width of crack, achieving acceptable CPU time and memory is a challenge when using the phase-field model to simulate the fracture problem, especially in three-dimensional modelling [18]. Consequently, many works focusing on improving the computational efficiency of phase-field model in different aspects were developed, such as parallel strategies [19, 20], adaptive methods [21-25], rapidly convergent algorithms [26, 27], and other methods [28-32]. Adaptive methods are rather popular because they not only significantly reduce the computational costs, but also avoid the conflict between crack prediction and mesh generation [33]. However, the majority of developed adaptive phase-field theories were implemented in two-dimensional problems, and only a few works concentrated on three-dimensional fracture [34, 35], which is more realistic and practical. In addition, the application of traditional multi-node elements is constrained by their order, which grows with the number of nodes and typically does not exceed quadratic order. To further improve the computing efficiency, this work adopts a type of hexahedral multi-node elements [36, 37], whose shape functions always take a trilinear form. Hence, elements with an arbitrary number of nodes can be used in numerical modelling. Moreover, a robust adaptive strategy for phase-field fracture model is proposed based on the trilinear multi-node elements.

The regularized internal length scale, which characterizes the width of the smeared localization band, has a significant influence on numerical results, especially for the peak force in the loading process [38]. Therefore, some works [39-41] took the internal length scale as a material property rather than a scale parameter. In order to resolve the problem caused by parameter selection, a length scale insensitive phase-field model is proposed by combining the phase-field regularized cohesive zone model (also called the unified phase-field model) [42-44]. Accurate results can be obtained as long as a reasonable relationship between mesh size and internal length scale is satisfied.

This paper is organized as follows: the formulations of the length scale insensitive phase-field model for three-dimensional brittle and quasi-brittle fracture are introduced in Section 2. In Section 3, the principle of trilinear hexahedral multi-node elements and the strategy of mesh refinement are provided. Section 4 gives the detailed finite element discretization and computer implementation for phase-field modelling. Some standard problems with various crack profiles are

solved to validate the capabilities of the proposed method in Section 5. Qualitative and quantitative comparisons with the standard phase-field model and physical experiment results are provided, as well. Finally, this work ends with conclusions in Section 6.

2. The length scale insensitive phase-field

2.1. Governing equations

Consider an elastic continuum Ω containing crack surfaces Γ_c , which is constrained by Dirichlet boundary condition $\bar{\mathbf{u}}$ and Neumann boundary condition $\bar{\mathbf{t}}$, as shown in **Fig. 1** (a). $\partial\Omega$ represents the external boundary of the domain Ω with the external normal vector $\mathbf{n}$. $\partial\Omega_f$ and $\partial\Omega_d$ are force boundary and displacement boundary, respectively.

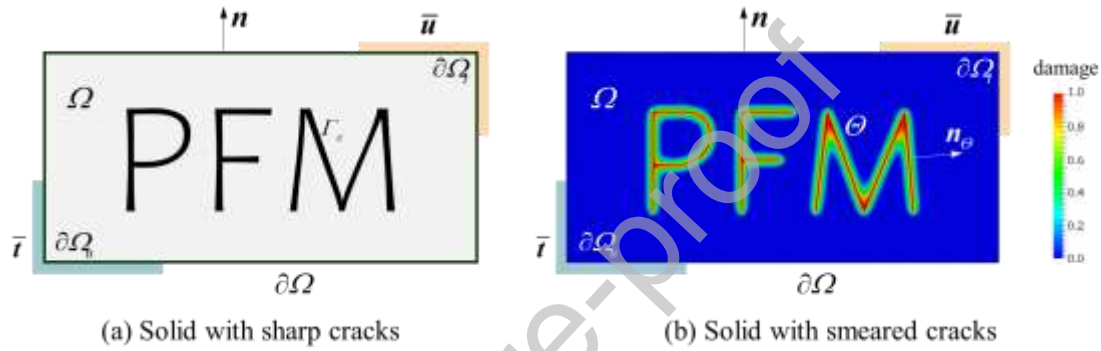


Fig. 1. A cracking solid with discrete and diffuse representations of the cracks.

In the framework of fracture mechanics, the energy functional for damage modelling can be expressed as

$$\Pi(\mathbf{u}, \Gamma_c) = \int_{\Omega} \psi_0(\boldsymbol{\varepsilon}(\mathbf{u})) d\Omega + G_c \int_{\Gamma_c} d\Gamma_c - \int_{\partial\Omega} \bar{\mathbf{t}} \cdot \mathbf{u} d\Omega \quad (1)$$

where G_c is the fracture toughness, and ψ_0 is the elastic strain energy density function which is computed by the strain tensor $\boldsymbol{\varepsilon}(\mathbf{u})$:

$$\psi_0 = \frac{1}{2} (\lambda \varepsilon_{kk} \varepsilon_{ii} + \mu \varepsilon_{ij} \varepsilon_{ij}) \quad (2)$$

where λ and μ are the Lamé constants, and the components of strain tensor are given as $\varepsilon_{ij} = 1/2 (u_{i,j} + u_{j,i})$.

In order to numerically track the crack evolution, the crack surface is regularized by introducing a crack surface density function, which can be written as

$$E_{\Gamma_c} = \int_{\Gamma_c} d\Gamma_c = \int_{\Theta} \gamma(s, \nabla s) d\Theta \quad (3)$$

where $\gamma(s, \nabla s)$ is the crack surface density function to smear the crack and given by

$$\gamma(s, \nabla s) = \frac{1}{c_0} \left[\frac{1}{l_0} \alpha(s) + l_0 |\nabla s|^2 \right] \quad (4)$$

where $s \in [0,1]$ is a continuous phase-field variable that characterizes the damage degree, ranging from 0 (intact region) to 1 (fully broken region). ∇s is the damage gradient that plays an important role in the following modelling of crack growth. l_0 is the regularized internal length scale measuring the width of smeared localization band Θ , as shown in **Fig. 1** (b). When the positive length scale is taken as $l_0 \rightarrow 0$, a sharp crack can be obtained. The scaling parameter $c_0 = 4 \int_0^1 \alpha(\beta) d\beta$ [45] is a constant ensuring the fracture toughness G_c be reproduced for complete failure state. The monotonically increasing crack geometric function $\alpha(s)$ needs to meet the conditions $\alpha(0) = 0$, $\alpha(1) = 1$, $\alpha'(s) \geq 0$, and a quadratic form can be considered

$$\alpha(s) = 2s - s^2 \geq 0 \quad \forall s \in [0,1] \quad (5)$$

in the work so that $c_0 = \pi$ can be obtained. Other forms of the crack geometric function, such as $\alpha(s) = s$ for the so called AT1 model [46] and $\alpha(s) = s^2$ for AT2 model [47], are also popular in phase-field models. Both models, however, are sensitive to the internal length scale l_0 , and AT2 model can result in an infinite damage profile. Moreover, some other regularized functions of crack surface energy are also usually adopted for phase-field model, such as the fourth-order type [48] and double-well type [40] functions. However, the double-well type function not only causes a confusion between intact and fully damaged states but also provides an energy barrier when the material tends to fracture, though the irreversibility of cracking can be modelled easily.

Correspondingly, the elastic potential energy is replaced by a modified form with a degradation function as

$$\int_{\Omega} \psi(\boldsymbol{\varepsilon}(\mathbf{u}), s) d\Omega = \int_{\Omega} \omega(s) \psi_0(\boldsymbol{\varepsilon}(\mathbf{u})) d\Omega \quad (6)$$

where $\psi(\boldsymbol{\varepsilon}(\mathbf{u}), s)$ is the degraded elastic potential energy, and the monotonically decreasing $\omega(s)$ is known as the degradation function which yields to the properties

$$\omega(0) = 1, \quad \omega(1) = 0, \quad \omega(1)' = 0 \quad (7)$$

Now, the total potential energy functional can be rewritten as

$$\Pi(\mathbf{u}, s) = \int_{\Omega} \omega(s) \psi_0(\boldsymbol{\varepsilon}(\mathbf{u})) d\Omega + \int_{\Theta} \gamma(s, \nabla s) d\Theta - \int_{\partial\Omega} \bar{\mathbf{t}} \cdot \mathbf{u} d\partial\Omega \quad (8)$$

The unknowns $(\mathbf{u}, s)$ is a stationary point of the energy functional $\Pi(\mathbf{u}, s)$, that is, the condition

$$\delta \Pi(\mathbf{u}, s) \geq 0 \quad (9)$$

should be satisfied for any admissible virtual displacement $\delta \mathbf{u} = \mathbf{0}$ and virtual phase-field $\delta s \geq 0$ [16]. The variation can also be rewritten in the expanded form

$$\begin{aligned} \delta \Pi(\mathbf{u}, s) = & \int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}, s) : \delta \boldsymbol{\varepsilon} d\Omega + \int_{\Omega} \frac{\partial \psi(\boldsymbol{\varepsilon}, s)}{\partial s} \delta s d\Omega \\ & + \int_{\Theta} G_c \left[\frac{\partial \gamma(s, \nabla s)}{\partial s} \delta s + \frac{\partial \gamma(s, \nabla s)}{\partial \nabla s} \cdot \delta(\nabla s) \right] d\Theta - \int_{\partial\Omega} \bar{\mathbf{t}} \cdot \delta \mathbf{u} d\partial\Omega \geq 0 \end{aligned} \quad (10)$$

By applying the divergence theorem to the given functional, the governing equations can be obtained in the forms:

$$\operatorname{div} \boldsymbol{\sigma}(\mathbf{u}, s) = 0 \quad \text{in } \Omega \quad (11)$$

and

$$\begin{cases} \omega'(s) \frac{\psi_0}{G_c} = \frac{2l_0}{c_0} \nabla^2 s - \frac{\alpha'(s)}{c_0 l_0} & \delta s > 0 \\ \omega'(s) \frac{\psi_0}{G_c} > \frac{2l_0}{c_0} \nabla^2 s - \frac{\alpha'(s)}{c_0 l_0} & \delta s = 0 \end{cases} \quad \text{in } \Theta \quad (12)$$

with the Neumann-type boundary conditions

$$\begin{cases} \boldsymbol{\sigma}(\mathbf{u}, s) \cdot \mathbf{n} = \bar{\mathbf{t}} & \text{on } \partial\Omega \\ \nabla s \cdot \mathbf{n}_{\Theta} = 0 & \text{on } \partial\Theta \end{cases} \quad (13)$$

The notations $\partial\Theta$ and $\mathbf{n}_{\Theta}$ are the boundary and external normal vectors of the smeared domain Θ , respectively; The degraded stress tensor $\boldsymbol{\sigma}(\mathbf{u}, s)$ is defined as

$$\boldsymbol{\sigma}(\mathbf{u}, s) = \frac{\partial \psi(\boldsymbol{\varepsilon}, s)}{\partial \boldsymbol{\varepsilon}} = \omega(s) \bar{\boldsymbol{\sigma}}(\mathbf{u}) \quad (14)$$

where $\bar{\boldsymbol{\sigma}}(\mathbf{u}) := \mathbb{D}_0 : \boldsymbol{\varepsilon}$ represents the effective stress tensor with the fourth-order elasticity tensor

$$\mathbb{D}_0 = \frac{E_0}{1 + \nu_0} (\mathbb{I} + \bar{\nu}_0 \mathbf{1} \otimes \mathbf{1}) \quad (15)$$

for the Young's modulus E_0 and Poisson's ratio ν_0 , and $\mathbf{1}$ as well as $\mathbb{I}$ are second-order and fourth-order unit tensors, respectively. $\bar{\nu}_0$ is a parameter related to ν_0 and can be decided by the stress state, i.e.,

$$\bar{\nu}_0 = \begin{cases} \nu_0 & \text{uniaxial stress} \\ \frac{\nu_0}{1 - \nu_0} & \text{plane stress} \\ \frac{\nu_0}{1 - 2\nu_0} & \text{other cases} \end{cases} \quad (16)$$

In order to prevent the crack from healing in the process of loading, a local history field with regard to the maximum tensile strain energy is considered as

$$\mathcal{H} = \max(h_{\min}, \max_{\tau \in [0, T]} h_{\tau}), \quad h_{\min} = \frac{1}{2E_0} f_t^2 \quad (17)$$

where $h_{\tau} = (1/2E_0) \bar{\boldsymbol{\sigma}}_{\tau}^2$ for the time step τ , and the appropriately postulated equivalent effective stress yields Rankine criterion as $\bar{\boldsymbol{\sigma}}_{\tau}(\boldsymbol{\sigma}) = \langle \bar{\boldsymbol{\sigma}}_{\tau} \rangle$; Macaulay brackets $\langle \cdot \rangle$ are defined as

$\langle x \rangle = \max(x, 0)$; f_t is the uniaxial tensile stress of the material; $\bar{\sigma}_1$ represents the first principal stress value of the effective stress tensor. The crack can only propagate in the domain where the local history field satisfies $\mathcal{H} \geq h_{\min}$, and the damage will develop accompanying the increase of $\mathcal{H}$.

Further, the governing equations Eq. (11) and Eq. (12) can be rewritten with the local history field as

$$\begin{cases} \operatorname{div} \boldsymbol{\sigma}(\mathbf{u}, s) = 0 & \text{in } \Omega \\ \omega'(s) \frac{\mathcal{H}}{G_c} = \frac{2l_0}{c_0} \nabla^2 s - \frac{\alpha'(s)}{c_0 l_0} & \text{in } \Theta \end{cases} \quad (18)$$

where the first formula is known as elastic equation for the displacement field, and the second is known as crack evolution equation for the phase-field.

Some other forms of local history field and degraded stress tensor can be found in [49-51]. Based on the energy decomposition, these models typically result in part of the strain energy being weakened. In this paper, the hybrid method proposed by Wu [42], in which the local history field is defined in an ad hoc manner, is adopted for the sake of simplicity.

2.2. Optimal degradation function

The degradation function plays a crucial role in mechanical responses and failure mode for phase-field model. In conventional phase-field models [46, 52, 53], the failure of materials is only characterized by a simple quadratic degradation function without any user-defined parameters, which is only applicable for brittle materials and induces length scale sensitive solutions. In order to avoid the issues, a class of new developed degradation functions, which makes the phase-field model be equivalent to a cohesive zone model [43] is applied in this work. The general formulation of the degradation functions is introduced as

$$\omega(s) := \frac{1}{1 + \varpi(s)} = \frac{(1-s)^p}{(1-s)^p + R(s)}, \quad \varpi(s) = \frac{R(s)}{(1-s)^p} \quad (19)$$

where the exponent is determined to be $p \geq 2$, and the continuous polynomial $R(s)$ takes the form

$$R(s) = b_1 s + b_1 b_2 s^2 + b_1 b_2 b_3 s^3 + \dots = b_1 s \cdot Q(s) \quad (20)$$

$$Q(s) = 1 + b_2 s + b_2 b_3 s^2 + \dots \quad (21)$$

For the sake of simplicity, the polynomial $Q(s)$ can be regarded as a quadratic function in this work, though a higher order can reduce the discrepancy at cost of more parameters. Then only the following coefficients require to be calibrated as

$$b_1 = \frac{4}{c_0 l_0} \frac{E_0 G_c}{f_t^2} \quad (22)$$

$$b_2 = \frac{1}{2} \left[\left(-\frac{16\pi}{c_0} \frac{G_c}{f_t^2} k_0 \right)^{\frac{2}{3}} + 1 \right] - (p+1) \quad (23)$$

$$b_3 = \begin{cases} 0 & p > 2 \\ \frac{1}{b_2} \left[\frac{1}{2} \left(\frac{c_0 w_c f_t}{2\pi G_c} \right)^2 - (1+b_2) \right] & p = 2 \end{cases} \quad (24)$$

where $k_0 < 0$ represents the initial slope of the target softening curve, and w_c is the ultimate crack opening with vanishing stress. The initial slope and ultimate crack opening can be computed as

$$k_0 = -\frac{c_0}{16\pi} \frac{f_t^2}{G_c} \left[2(b_2 + p + 1) - 1 \right]^{3/2} \quad (25)$$

$$w_c = \frac{2\pi G_c}{c_0 f_t} \sqrt{2Q(1)} \lim_{s^* \rightarrow 1} (1-s^*)^{1-p/2} \quad (26)$$

where s^* is the maximum value of s .

Therefore, once the standard material properties and the target softening law are given, the parameters in $Q(s)$ and the final degradation function can be determined. It is notable that b_2 and b_3 are independent of the internal length scale l_0 , and all the coefficients are controlled by the tensile strength f_t , that is, the stress state is insensitive to the smeared width of crack.

By using the proper softening law, the optimal degradation function can be obtained for various brittle and quasi-brittle solids. For example, the following frequently adopted softening curves are recommended,

- Linear softening curve

$$k_0 = -\frac{f_t^2}{2G_c}, \quad w_c = \frac{2G_c}{f_t}, \quad p = 2 \quad \Rightarrow \quad b_2 = -\frac{1}{2}, \quad b_3 = 0 \quad (27)$$

- Exponential softening curve

$$k_0 = -\frac{f_t^2}{G_c}, \quad w_c = +\infty, \quad p = 2.5 \quad \Rightarrow \quad b_2 = 0.1748, \quad b_3 = 0 \quad (28)$$

- Cornelissen's softening curve

$$k_0 = -1.3546 \frac{f_t^2}{G_c}, \quad w_c = 5.1361 \frac{G_c}{f_t}, \quad p = 2 \quad \Rightarrow \quad b_2 = 1.3868, \quad b_3 = 0.6567 \quad (29)$$

Note that the Cornelissen's [54] softening curve, calibrated from physical experiments, is often applied in the modelling of concrete. Besides, some other popular softening curves can also be considered; see [55].

Remark 2.1. Some other regularized functions of crack surface energy are also usually adopted

for phase-field model, such as the fourth-order type [48] and double-well type [40] functions. However, the double-well type function not only causes a confusion between intact and fully damaged states but also provides an energy barrier when the material tends to fracture, though the irreversibility of cracking can be modelled easily.

3. Mesh adaptive technique using trilinear multi-node transition elements

In this section, an efficient adaptive strategy for phase-field fracture model is introduced. Although there are many studies [56-58] have assessed how to resolve the problem of computational costs through adaptive methods, most of them concentrated on two-dimensional problems.

Compared with the traditional elements, this type of element replaces the polynomial shape functions with a trilinear form. Hence, a constant order can be kept regardless of the number of nodes in the element, which is a substantial enhancement for the application of mesh adaptation.

The trilinear multi-node transition elements [37] are adopted in this paper for mesh refinement. Compared with the traditional elements [59], this type of element replaces the polynomial shape functions with a trilinear form. Hence, a constant order can be kept regardless of the number of nodes in the element, which is a substantial improvement for the application of mesh adaptation. Moreover, a concise adaptive criterion is outlined to preserve numerical precision.

3.1. Shape functions of trilinear elements

The shape functions of trilinear-type multi-node transition elements can be derived by applying the moving least square (MLS) method. Let $\mathbf{u}(\xi)$ be a 3-D vector field with three mutually perpendicular components $u_i (i = 1, 2, 3)$. Correspondingly, the master coordinate ξ can be expressed using its components (ξ_1, ξ_2, ξ_3) , indicated by (ξ, η, ζ) for convenience.

Suppose $\bar{\xi}$ is the center of a spherical domain, one can take a local MLS approximation $\mathbf{u}^h(\xi, \bar{\xi})$ for $\mathbf{u}(\xi)$ based on m polynomials as

$$\mathbf{u}^h(\xi, \bar{\xi}) = \sum_{i=1}^m a_i(\bar{\xi}) p_i(\xi - \bar{\xi}) = \mathbf{a}^T(\bar{\xi}) \mathbf{p}(\xi - \bar{\xi}) \quad (30)$$

where $\{a_i(\bar{\xi})\}_{i=1}^m$ is a set of unknown coefficients depending on the point $\bar{\xi}$, which constitutes the $3 \times m$ matrix $\mathbf{a}^T(\bar{\xi})$. $\mathbf{p}(\xi - \bar{\xi})$ is a $1 \times m$ matrix consisting of the polynomial basis functions $p_i(\xi - \bar{\xi})$ and can be written as

$$\mathbf{p}^T(\xi - \bar{\xi}) = [1, \xi - \bar{\xi}, \eta - \bar{\eta}, \zeta - \bar{\zeta}, (\xi - \bar{\xi})(\eta - \bar{\eta}), \dots] \quad (31)$$

For n nodal points in the domain, the error functional of the approximation $\mathbf{u}^h(\xi, \bar{\xi})$ in the least square sense is donated by

$$J = \sum_{j=1}^n w_j(\xi - \bar{\xi}) \left[\mathbf{a}^T(\bar{\xi}) \mathbf{p}(\xi_j - \bar{\xi}) - \mathbf{u}_j \right]^2 \quad (32)$$

where w_j and $\mathbf{u}_j$ are the weight function and the nodal value for the node $j(j=1, 2, \dots, n)$.

The error functional Eq. (32) can also be rewritten in a matrix form as

$$\mathbf{J} = [\mathbf{P}\mathbf{a} - \mathbf{U}]\mathbf{W}[\mathbf{P}\mathbf{a} - \mathbf{U}] \quad (33)$$

where

$$\mathbf{P} = \mathbf{P}(\xi_j - \bar{\xi}) = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ (\xi_1 - \bar{\xi}) & (\xi_2 - \bar{\xi}) & (\xi_3 - \bar{\xi}) & \dots & (\xi_n - \bar{\xi}) \\ (\eta_1 - \bar{\eta}) & (\eta_2 - \bar{\eta}) & (\eta_3 - \bar{\eta}) & \dots & (\eta_n - \bar{\eta}) \\ (\zeta_1 - \bar{\zeta}) & (\zeta_2 - \bar{\zeta}) & (\zeta_3 - \bar{\zeta}) & \dots & (\zeta_n - \bar{\zeta}) \\ (\xi_1 - \bar{\xi})(\eta_1 - \bar{\eta}) & (\xi_2 - \bar{\xi})(\eta_2 - \bar{\eta}) & (\xi_3 - \bar{\xi})(\eta_3 - \bar{\eta}) & \dots & (\xi_n - \bar{\xi})(\eta_n - \bar{\eta}) \\ (\eta_1 - \bar{\eta})(\zeta_1 - \bar{\zeta}) & (\eta_2 - \bar{\eta})(\zeta_2 - \bar{\zeta}) & (\eta_3 - \bar{\eta})(\zeta_3 - \bar{\zeta}) & \dots & (\eta_n - \bar{\eta})(\zeta_n - \bar{\zeta}) \\ (\zeta_1 - \bar{\zeta})(\xi_1 - \bar{\xi}) & (\zeta_2 - \bar{\zeta})(\xi_2 - \bar{\xi}) & (\zeta_3 - \bar{\zeta})(\xi_3 - \bar{\xi}) & \dots & (\zeta_n - \bar{\zeta})(\xi_n - \bar{\xi}) \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}_{m \times n} \quad (34)$$

$$\mathbf{W} = \mathbf{W}(\xi_j - \bar{\xi}) = \begin{bmatrix} w_1(\xi_1 - \bar{\xi}) & 0 & 0 & \dots & 0 \\ 0 & w_2(\xi_2 - \bar{\xi}) & 0 & \dots & 0 \\ 0 & 0 & w_3(\xi_3 - \bar{\xi}) & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & w_n(\xi_n - \bar{\xi}) \end{bmatrix}_{n \times n} \quad (35)$$

$$\mathbf{U}^T = \begin{bmatrix} u_{11} & u_{12} & u_{13} & \dots & u_{1n} \\ u_{21} & u_{22} & u_{23} & \dots & u_{2n} \\ u_{31} & u_{32} & u_{33} & \dots & u_{3n} \end{bmatrix}_{3 \times n} \quad (36)$$

By minimizing the functional Eq. (33), the derivative of J in terms of $\mathbf{a}(\bar{\xi})$ can be obtained as

$$\frac{\partial \mathbf{J}}{\partial \mathbf{a}} = \mathbf{M}(\bar{\xi})\mathbf{a}(\bar{\xi}) - \mathbf{B}(\bar{\xi})\mathbf{U} = \mathbf{0} \quad (37)$$

with

$$\mathbf{M} = \mathbf{P}\mathbf{W}\mathbf{P}^T \quad \text{and} \quad \mathbf{B} = \mathbf{P}\mathbf{W} \quad (38)$$

Combining Eq. (30) with Eq. (37), the local approximation $\mathbf{u}^h(\xi, \bar{\xi})$ can be written as

$$\mathbf{u}^h(\xi, \bar{\xi}) = \mathbf{U}^T \mathbf{B}^T(\bar{\xi}) \mathbf{M}^{-1}(\bar{\xi}) \mathbf{p}(\xi - \bar{\xi}) = \sum_{j=1}^n \phi_j(\xi, \bar{\xi}) \mathbf{u}_j \quad (39)$$

Then from the moving process $\lim_{\xi \rightarrow \bar{\xi}} \mathbf{u}^h(\xi, \bar{\xi}) = \mathbf{u}^h(\bar{\xi})$, one can have

$$\lim_{\xi \rightarrow \bar{\xi}} \mathbf{p}^T(\xi - \bar{\xi}) = \mathbf{p}^T(\mathbf{0}) = [1, 0, 0, 0, 0, \dots] \quad (40)$$

and

$$\mathbf{u}^h(\xi) = \mathbf{U}^T \mathbf{B}^T(\xi) \mathbf{M}^{-1} \mathbf{p}(\mathbf{0}) = \mathbf{U}^T \boldsymbol{\phi}(\xi) = \sum_{j=1}^n \phi_j(\xi) \mathbf{u}_j \quad (41)$$

where $\boldsymbol{\phi}(\xi) = \mathbf{B}^T(\xi) \mathbf{M}^{-1} \mathbf{p}(\mathbf{0}) = \mathbf{P}^{-1}(\xi) \mathbf{p}(\mathbf{0})$ represents the shape functions with its components $\{\phi_j(\xi)\}_{j=1}^n$ defined as

$$\phi_j(\xi) = \sum_{k=1}^m \sum_{l=1}^m B_{kj}(\xi) M_{kl}^{-1}(\xi) \mathbf{p}_l(\mathbf{0}) \quad (42)$$

In order to be applicable in the finite element method, the shape functions must satisfy the following conditions:

$$\text{Partition of unity: } \sum_{j=1}^n \phi_j(\xi) = 1 \quad (43)$$

$$\text{Linear completeness: } \sum_{j=1}^n \phi_j(\xi) \xi_j = \xi \quad (44)$$

$$\text{Kronecker-delta property: } \phi_j(\xi_k) = \begin{cases} 1 & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases} \quad (k=1, 2, \dots, j, \dots, n) \quad (45)$$

$$\text{Disappear on edges not contain the node: } \phi_j(\xi_{pe}) = 0 \quad (46)$$

where ξ_{pe} is the arbitrary location on the edges not meeting with the node “j”.

Due to the included complete linear polynomial in $\mathbf{p}(\xi - \bar{\xi})$, the completeness conditions Eq. (43) and Eq. (44) can be satisfied naturally by the MLS-based shape functions in Eq. (42). Besides, the matrix $\mathbf{M}(\xi)$ is supposed to be invertible and well-conditioned due to the existence of $\phi_j(\xi)$, that is, the rank of $\mathbf{W}(\xi)$ is greater than or equal to n . Therefore, for a hexahedral element with eight nodes, one can presume that $\text{Rank}(\mathbf{W}) = 8$ and donate $\mathbf{p}(\xi)$ as

$$\mathbf{p}(\xi) = [1 \quad \xi \quad \eta \quad \zeta \quad \xi\eta \quad \eta\zeta \quad \zeta\xi \quad \xi\eta\zeta] \quad (47)$$

so that the C^1 continuity in the element with the compatibility conditions Eqs. (45)-(46) on the boundary can be fulfilled.

Presuming that the hexahedral element corresponds to a square domain with $-1 \leq \xi, \eta, \zeta \leq 1$, the trilinear shape functions can be obtained by substituting the mentioned conditions Eqs. (43)-(46) and the polynomial basis Eq. (47) into Eq. (42) as

$$\phi_j(\xi) = \frac{1}{8} (1 + \xi_j \xi) (1 + \eta_j \eta) (1 + \zeta_j \zeta) \quad (48)$$

which is commonly used in conventional eight-node elements.

For the elements with more than eight nodes, extra trilinear basis functions can be added in $\mathbf{p}(\xi)$ to meet the interpolation property. Suppose that there are p , q and r nodes inserted respectively on the edges paralleling with ξ , η and ζ directions, then pq , qr , and rp nodes should be placed on corresponding faces. For instance, consider an element with

$N = 8 + 3p + 3q + 3r + pq + qr + rp$ ($p, q, r = 2, 1, 1$) nodes as shown in **Fig. 2**, the polynomial basis functions can be written as

$$\begin{aligned}
 \mathbf{p}(\xi) = & [1, \xi, \eta, \zeta, \xi\eta, \eta\zeta, \zeta\xi, \xi\eta\zeta, \\
 & |\xi - \xi_9|(\eta + \text{sign}(\eta_9))(\zeta + \text{sign}(\zeta_9)), \dots, |\xi - \xi_{8+3p}|(\eta + \text{sign}(\eta_{8+3p}))(\zeta + \text{sign}(\zeta_{8+3p})), \\
 & |\eta - \eta_{8+3p+1}|(\xi + \text{sign}(\xi_{8+3p+1}))(\zeta + \text{sign}(\zeta_{8+3p+1})), \dots, |\eta - \eta_{8+3p+3q}|(\xi + \text{sign}(\xi_{8+3p+3q}))(\zeta + \text{sign}(\zeta_{8+3p+3q})), \\
 & |\zeta - \zeta_{8+3p+3q+1}|(\xi + \text{sign}(\xi_{8+3p+3q+1}))(\eta + \text{sign}(\eta_{8+3p+3q+1})), \dots, |\zeta - \zeta_{8+3p+3q+3r}|(\xi + \text{sign}(\xi_{8+3p+3q+3r}))(\eta + \text{sign}(\eta_{8+3p+3q+3r})), \\
 & |\xi - \xi_{8+3p+3q+3r+1}|(\eta + \text{sign}(\eta_{8+3p+3q+3r+1}))(\zeta + \text{sign}(\zeta_{8+3p+3q+3r+1})), \dots, |\xi - \xi_{8+3p+3q+3r+pq}|(\eta + \text{sign}(\eta_{8+3p+3q+3r+pq}))(\zeta + \text{sign}(\zeta_{8+3p+3q+3r+pq})), \\
 & |\eta - \eta_{8+3p+3q+3r+pq+1}|(\xi + \text{sign}(\xi_{8+3p+3q+3r+pq+1}))(\zeta + \text{sign}(\zeta_{8+3p+3q+3r+pq+1})), \dots, \\
 & |\eta - \eta_{8+3p+3q+3r+pq+qr}|(\xi + \text{sign}(\xi_{8+3p+3q+3r+pq+qr}))(\zeta + \text{sign}(\zeta_{8+3p+3q+3r+pq+qr})), \\
 & |\zeta - \zeta_{8+3p+3q+3r+pq+qr+1}|(\xi + \text{sign}(\xi_{8+3p+3q+3r+pq+qr+1}))(\eta + \text{sign}(\eta_{8+3p+3q+3r+pq+qr+1})), \dots, \\
 & |\xi - \xi_{8+3p+3q+3r+pq+qr+pr}|(\eta + \text{sign}(\eta_{8+3p+3q+3r+pq+qr+pr}))(\zeta + \text{sign}(\zeta_{8+3p+3q+3r+pq+qr+pr}))] \quad (49) \\
 = & [1, \xi, \eta, \zeta, \xi\eta, \eta\zeta, \zeta\xi, \xi\eta\zeta, \\
 & |\xi - \xi_9|(\eta + \text{sign}(\eta_9))(\zeta + \text{sign}(\zeta_9)), \dots, |\xi - \xi_{14}|(\eta + \text{sign}(\eta_{14}))(\zeta + \text{sign}(\zeta_{14})), \\
 & |\eta - \eta_{15}|(\xi + \text{sign}(\xi_{15}))(\zeta + \text{sign}(\zeta_{15})), \dots, |\eta - \eta_{17}|(\xi + \text{sign}(\xi_{17}))(\zeta + \text{sign}(\zeta_{17})), \\
 & |\zeta - \zeta_{18}|(\xi + \text{sign}(\xi_{18}))(\eta + \text{sign}(\eta_{18})), \dots, |\zeta - \zeta_{20}|(\xi + \text{sign}(\xi_{20}))(\eta + \text{sign}(\eta_{20})), \\
 & |\xi - \xi_{21}|(\eta + \text{sign}(\eta_{21}))(\zeta + \text{sign}(\zeta_{21})), |\xi - \xi_{22}|(\eta + \text{sign}(\eta_{22}))(\zeta + \text{sign}(\zeta_{22})), \\
 & |\eta - \eta_{23}|(\xi + \text{sign}(\xi_{23})), \\
 & |\xi - \xi_{24}|(\eta + \text{sign}(\eta_{24})), \dots, |\xi - \xi_{25}|(\eta + \text{sign}(\eta_{25}))]
 \end{aligned}$$

Consequently, all the shape functions can be established in a trilinear form by employing Eq. (42). In contrast to the multi-node elements used frequently in other works, this kind of element can prevent the rising of the order of shape functions.

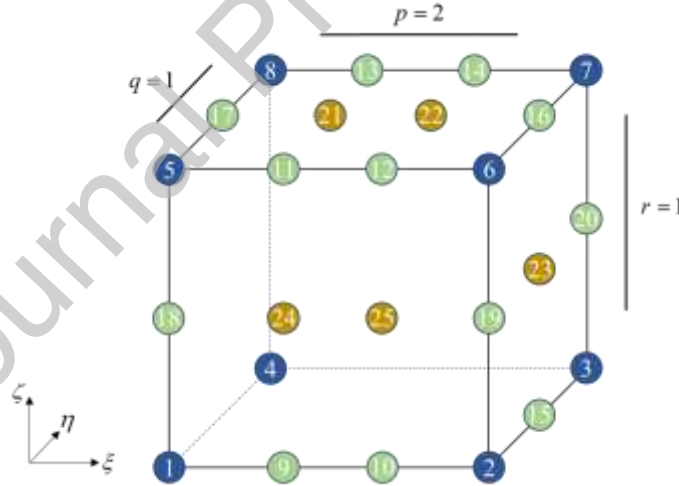


Fig. 2. An element with $N=8+3p+3q+3r+pq+qr+rp$ nodes ($p=2, q=1, r=1$, nodes on corners rendered blue, edges rendered green, and faces rendered yellow).

In this work, the number of inserted nodes is confined to

$$p \leq 1, \quad q \leq 1, \quad r \leq 1 \quad (50)$$

in an element for the sake of simplicity. Take a fourteen-node element illustrated in **Fig. 3(a)**, which may be generated during mesh refinement, as an example, the shape functions for each

node can be computed easily, as plotted in **Fig. 3(b)**. It is evident that the necessary properties of the shape functions, described by Eqs. (45)-(46), are satisfied for all nodes.

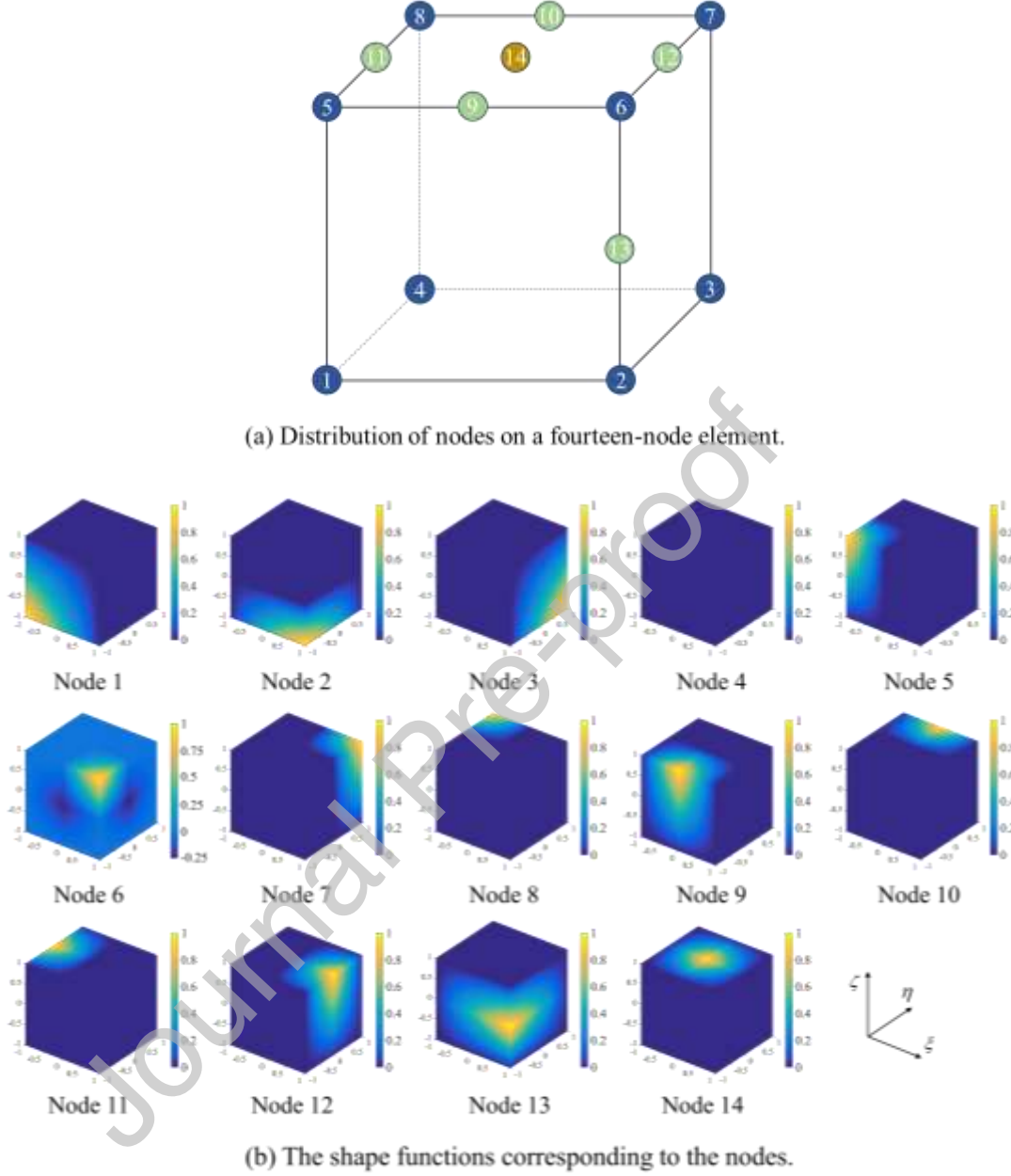


Fig. 3. The position and shape function of nodes for a fourteen-node element.

Furthermore, a special distribution strategy of integration points is required to satisfy the C^1 continuity, more details are referred to **Remark 3.1**.

Remark 3.1. In a trilinear-type element, the support of individual shape function is limited to a part of the element domain and makes no sense for the rest region, as shown in **Fig. 3**. This feature accounts for a relaxation of the C^1 continuity on the interfaces of subdomains in the element, as depicted in **Fig. 4**. The interfaces which partition the multi-node element into $(p+1)(q+1)(r+1)$

subdomains, are determined according to the location of nodes on the element edges.

However, the requirement of C^1 continuity along the interfaces can be removed by considering the discontinuity of derivatives in the numerical integration. In order to solve the problem, one can distribute $2 \times 2 \times 2$ Gauss integration points in all subdomains. Therefore, the C^1 continuity is only required inside the subdomains.

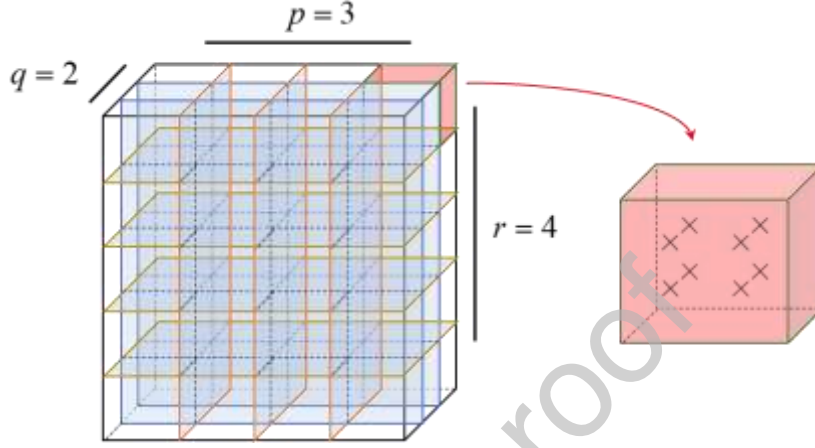


Fig. 4. An example of subdomains for the Gauss integration ($p=3$, $q=2$, $r=4$, and 60 subdomains are constructed).

Remark 3.2. Because the C^1 continuity is relaxed on the interfaces of the subdomains, the trilinear-type elements can also be used to connect non-matching meshes while traditional multi-node elements do not possess the capacity.

3.2. Mesh refinement and adaptive criterion.

During the refinement process, as shown in **Fig. 5**, one parental element, which is at a lower level, can be divided into eight subelements. Generally, the mesh refinement is only carried out in the region where the crack initiates and propagates. Moreover, the level gap between the neighbor elements is limited to one at most, so that a high precision can be kept.

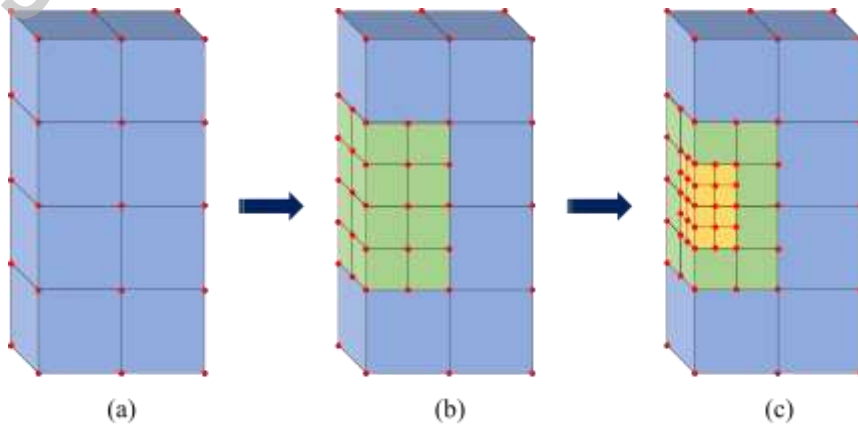


Fig. 5. Generation of quadrature subelements and nodes for regular multi-node hexahedron

elements: (a) before refinement, (b) the first refinement step, and (c) the second refinement step.

Although some adaptive algorithms have been proposed for phase-field model, most of them were established only based on the phase-field variable and did not take the original mesh size into account [18, 60]. This may lead to an irrational refinement for nonuniform mesh, in which the proper relationship between the mesh size and internal length scale is difficult to be reached for every element. Thus, a robust adaptive criterion determined by the phase-field variable and the element size, is used to guarantee both efficiency and accuracy in this work.

For an element with the size d and the phase-field value s_e , the adaptive criterion is triggered when the relationship is satisfied

$$d > \tilde{d} \quad (51)$$

where the element size is equal to its longest edge, or other characteristic length can also be applied. The parameter $\tilde{d}$ represents the critical element size which is given as

$$\tilde{d} = \begin{cases} d_0 & \text{if } s_e \leq s_{\min} \\ \frac{(s_{\max} - s_e) \times (d_0 - l_0 / c)}{s_{\max} - s_{\min}} + l_0 / c & \text{if } s_{\min} \leq s_e \leq s_{\max} \\ l_0 / c & \text{if } s_e \geq s_{\max} \end{cases} \quad (52)$$

where d_0 represents the initial mesh size, and c is the ratio between the internal length scale l_0 and required minimum element size $h_{\min}$ in the damage domain. An element will be divided into subelements when its phase-field value s_e is bigger than $s_{\min}$ and Eq. (51) is satisfied, otherwise it will keep the current state.

Notably, inappropriate values for the parameters may result in a certain amount of error, especially in the aspect of force response. The ratio c has little influence on the numerical accuracy so long as a proper internal length scale l_0 is applied and the condition $c \geq 2$ is satisfied. However, the numerical result is sensitive to the choice of adaptive phase-field range $[s_{\min}, s_{\max}]$, although it can be ignored when sufficiently small values are taken. Considering the balance between efficiency and precision, the values of the parameters are recommended as

$$c = 4.0, \quad s_{\min} = 0.01, \quad s_{\max} = 0.1 \quad (53)$$

Undoubtedly, other values can also be considered according to personal demands. If larger values of $s_{\min}$ and $s_{\max}$ are taken, the computational efficiency can be improved with a loss of precision. More details will be discussed in Section 5.1.

4. Numerical implementation

Combined with the efficient adaptive strategy, the three-dimensional phase-field model can be

implemented readily. For the adaptive length scale insensitive phase-field model, several parameters are required for numerical implementation, i.e., Young's modulus E_0 , Poisson's ratio ν_0 , fracture toughness G_c , uniaxial tensile strength f_t , the internal length scale l_0 , the adaptive ratio c , and the adaptive phase-field range $[s_{\min}, s_{\max}]$. In the light of the recommended values in Eq. (53), the numerical error caused by the adaptive parameters can be neglected, which will be proved in Section 5.

In the framework of FEM, the discrete form of governing equations Eq. (18) can be written as

$$\int_{\Omega} \mathbf{B}_u^T \mathbf{D} \mathbf{B}_u d\Omega \hat{\mathbf{u}} = \int_{\Omega} \Phi_u^T \bar{\mathbf{t}} d\Omega \quad (54)$$

$$\int_{\Theta} \frac{2l_0}{c_0} \mathbf{B}_s^T \mathbf{B}_s d\Theta \hat{\mathbf{s}} + \int_{\Theta} \frac{1}{c_0 l_0} \alpha'(s) \Phi_s^T d\Theta + \int_{\Theta} \omega'(s) \mathcal{H}(\Phi_s^T) d\Theta = \mathbf{0} \quad (55)$$

where $\hat{\mathbf{u}}$ and $\hat{\mathbf{s}}$ represent the nodal displacement and nodal phase-field, respectively. $\mathbf{D}$ is the degraded constitutive matrix which is given as

$$\mathbf{D} := \frac{\partial \sigma(\mathbf{u}, s)}{\partial \epsilon} = \omega(s) \mathbb{D}_0 \quad (56)$$

Besides, for a point with the coordinate $\mathbf{x} = (x, y, z)$ in the global coordinate system, the interpolation matrices for displacement field and phase-field are

$$\Phi_u(\mathbf{x}) = [\phi_1 \mathbf{I} \quad \phi_2 \mathbf{I} \quad \cdots \quad \phi_N \mathbf{I}] \quad (57)$$

$$\Phi_s(\mathbf{x}) = [\phi_1 \quad \phi_2 \quad \cdots \quad \phi_N] \quad (58)$$

respectively. N is the number of nodes in the element, and $\mathbf{I}$ is the identity matrix.

The compatibility matrices are given as

$$\mathbf{B}_u(\mathbf{x}) = [\mathbf{B}_u^1 \quad \mathbf{B}_u^2 \quad \cdots \quad \mathbf{B}_u^N] \quad (59)$$

$$\mathbf{B}_s(\mathbf{x}) = [\mathbf{B}_s^1 \quad \mathbf{B}_s^2 \quad \cdots \quad \mathbf{B}_s^N] \quad (60)$$

with the components

$$\mathbf{B}_u^i = \begin{bmatrix} \frac{\partial \phi_i}{\partial x} & 0 & 0 & \frac{\partial \phi_i}{\partial y} & 0 & \frac{\partial \phi_i}{\partial z} \\ 0 & \frac{\partial \phi_i}{\partial y} & 0 & \frac{\partial \phi_i}{\partial x} & \frac{\partial \phi_i}{\partial z} & 0 \\ 0 & 0 & \frac{\partial \phi_i}{\partial z} & 0 & \frac{\partial \phi_i}{\partial y} & \frac{\partial \phi_i}{\partial x} \end{bmatrix}^T \quad (61)$$

$$\mathbf{B}_s^i = \begin{bmatrix} \frac{\partial \phi_i}{\partial x} & \frac{\partial \phi_i}{\partial y} & \frac{\partial \phi_i}{\partial z} \end{bmatrix}^T \quad (62)$$

There are two frequently adopted solving strategies for phase-field model, that is, the monolithic and staggered schemes. In this work, the staggered scheme is applied. The whole loading process can be discretized into a certain number of substeps, wherein the equations (54) and (55) are solved sequentially.

In each substep, the adaptive criterion is used after solving the displacement field and phase-field. Generally, the mesh refinement will be carried out if there is any element that satisfies the adaptive criterion Eq. (51). The governing equations are supposed to be solved again in the substep to ensure a reliable result if the mesh refinement process is performed. More details about the adaptive algorithm can be found in **Algorithm 1**.

Algorithm 1: Solution algorithm for TAPFM

```

for every successive time step  $n+1$  do
    Initialization:  $(\hat{\mathbf{u}}_{n+1}^{(old)}, \hat{\mathbf{s}}_{n+1}^{(old)}) = (\hat{\mathbf{u}}_n, \hat{\mathbf{s}}_n)$ 
    do
        Compute nodal displacements  $\hat{\mathbf{u}}_{n+1}^{(new)}$  with fixed phase-field  $\hat{\mathbf{s}}_{n+1}^{(old)}$ 

$$\int_{\Omega} \mathbf{B}_u^T \mathbf{D} \mathbf{B}_u d\Omega \hat{\mathbf{u}}_{n+1}^{(new)} = \int_{\Omega} \mathbf{\Phi}_u^T \bar{\mathbf{t}} d\Omega, \quad \mathbf{D} = \omega(\hat{\mathbf{s}}_{n+1}^{(old)}) \mathbb{D}_0$$

        Compute phase-field dofs  $\hat{\mathbf{s}}_{n+1}^{(new)}$  with fixed displacement  $\hat{\mathbf{u}}_{n+1}^{(new)}$ 

$$\int_{\Theta} \frac{2l_0}{c_0} \mathbf{B}_s^T \mathbf{B}_s d\Theta \hat{\mathbf{s}}_{n+1}^{(new)} + \int_{\Theta} \frac{1}{c_0 l_0} \alpha'(\hat{\mathbf{s}}_{n+1}^{(new)}) \mathbf{\Phi}_s^T d\Theta + \int_{\Theta} \omega'(\hat{\mathbf{s}}_{n+1}^{(new)}) \mathcal{H} \mathbf{\Phi}_s^T d\Theta = \mathbf{0}, \quad \mathcal{H} = \mathcal{H}(\hat{\mathbf{u}}_{n+1}^{(new)})$$

        subjected to:  $0 \leq \hat{\mathbf{s}}_{In} \leq \hat{\mathbf{s}}_{In+1}^{(new)} \leq 1, \quad \forall I = 1, 2, \dots, N$ 
        Find and refine the elements with  $\hat{\mathbf{u}}_{n+1}^{(new)}$  and  $\hat{\mathbf{s}}_{n+1}^{(new)}$ : Algorithm 2
    until no element is refined
    Update nodal unknowns:  $(\hat{\mathbf{u}}_{n+1}, \hat{\mathbf{s}}_{n+1}) = (\hat{\mathbf{u}}_{n+1}^{(new)}, \hat{\mathbf{s}}_{n+1}^{(new)})$ 
end

```

For the refinement of the mesh, an element-wise approach is adopted. At every substep, an element set, Y is used to indicate the elements to be refined. On the solution domain Ω , if the size d of an element e is greater than the critical value $\tilde{d}$, it is marked for refinement. To ensure that the level difference between any two adjacent elements is at most one, all the neighbors of the element $e \in Y$, whose level is bigger than e , are also chosen as elements supposed to be refined and added to Y . Then, new nodes are inserted on elements in Y , and

their variables can be interpolated using other nodes. Besides, some of the new nodes also need to be inserted on the elements which share edges with elements in Y . It should be noted that, the number and location of Gauss integration points in trilinear adaptive elements, should be reset according to the distribution of nodes. Finally, every coarser element in Y is divided into several children elements, which are one level higher than itself. For further clarity on the implementation of the adaptive refinement scheme, a step-by-step procedure is outlined in **Algorithm 2**.

Algorithm 2: Mesh refinement

Input: $\hat{\mathbf{u}}_{n+1}^{(old)}$, $\hat{\mathbf{s}}_{n+1}^{(old)}$, $\hat{\mathbf{u}}_{n+1}^{(new)}$, $\hat{\mathbf{s}}_{n+1}^{(new)}$

Set: X : total number of elements, $Y = \emptyset$: empty set to contain the elements need to be refined

for $i = 1, 2, \dots, X$ **do**

 Compute the critical element size for the element e_i

$$\tilde{d}_{e_i} = \tilde{d}_{e_i}(\hat{\mathbf{s}}_{n+1}^{(new)})$$

if $d_{e_i} > \tilde{d}_{e_i}$ **then**

 Add e_i into Y

 Find the neighbor elements, Ne_i of e_i

if $Ne_i.level < e_i.level$ **then**

 Add Ne_i into Y

end

end

end

if $Y \neq \emptyset$ **then**

 Insert nodes on the elements in Y and their neighbor elements

 Get $\hat{\mathbf{u}}_{n+1}^{(old)}$, $\hat{\mathbf{s}}_{n+1}^{(old)}$, $\hat{\mathbf{u}}_{n+1}^{(new)}$, $\hat{\mathbf{s}}_{n+1}^{(new)}$ on new nodes by interpolation

 Modify the Gauss integration points according to the distribution of nodes

 Divide the elements in Y

end

Output: Updated $\hat{\mathbf{u}}_{n+1}^{(old)}$, $\hat{\mathbf{s}}_{n+1}^{(old)}$, $\hat{\mathbf{u}}_{n+1}^{(new)}$, $\hat{\mathbf{s}}_{n+1}^{(new)}$

5. Numerical tests

In this section, several representative numerical examples are presented to demonstrate the

capabilities of the outlined method for brittle and quasi-brittle materials. In the beginning, the influence of the internal length scale l_0 , as well as the adaptive parameters on the force-displacement response in the proposed trilinear element-based adaptive phase-field model (TAPFM), is discussed in Section 5.1 by changing the parameters. Thus, in the first example, a single-edge notched plate under tensile or shear force is computed to verify the insensitivity of the model to the parameters. Besides, the standard nonadaptive phase-field model (PFM) is implemented as a comparison. Then, several bending tests with different fracture modes are reproduced to further exhibit the efficiency of the adaptive algorithm in Section 5.2. Applying different shape of initial defects and loading states on the beams, the mode-I, I + II, and I + III mode failures can be observed, respectively. Section 5.3 presents the application of the adaptive method to simulate the crack propagation under torsion. The final example in Section 5.4 concentrates on the coalescence and bifurcation of multi-crack problems.

If there is no special statement, the default values of the adaptive parameters in Eq. (53) are adopted in all examples. The proposed formulation is implemented in our in-house FE code based on the open-source library SuiteSparse. The computations are completed on the computer with Intel(R) Core(TM) i7-9750H CPU @ 2.60GHz and 16.0GB RAM.

5.1. Single-edge notched plate under tension or shear.

A square plate containing an edge flaw under mode I/II loading, similarly to the example presented by Miehe et al. [53], is modelled to simulate the fracture behavior in this section. As shown in **Fig. 6**, the side length of the square plate is 1mm, and the thickness is taken as 0.025mm. The initial horizontal flaw is 0.5mm long extending from the left edge to the plate's center. The material parameters are [43, 61]: Young's modulus $E_0 = 210$ GPa, Poisson's ratio $\nu_0 = 0.3$, failure tension strength $f_t = 2.445$ GPa, and fracture toughness $G_c = 0.0027$ kN/mm. Besides, the linear softening curve is assumed in the simulation.

5.1.1. Single-edge notched plate under mode-I loading.

As illustrated in **Fig. 6** (a), the plate is subjected to a vertically upward displacement on the top face. The constant incremental displacement is applied as $\Delta u_y^* = 2.0 \times 10^{-6}$ mm and the total displacement is $u^* = 0.006$ mm. Moreover, the bottom face is fixed in three directions. In **Fig. 6** (b), the domain is discretized by $40 \times 40 \times 1$ eight-node hexahedral finite elements, and the initial flaw is set as a discontinuous surface in the modelling.

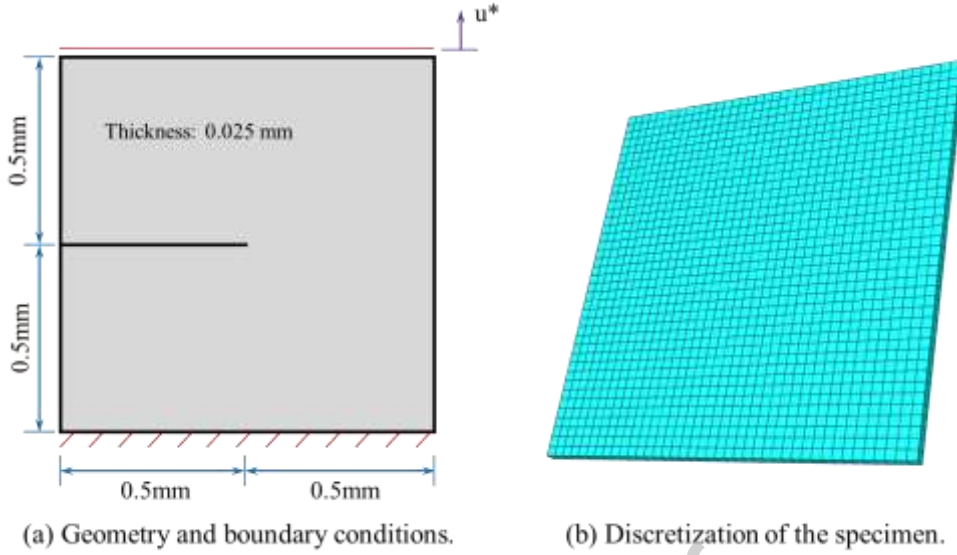


Fig. 6. Single-edge notched plate under mode-I loading: (a) Geometry and boundary conditions and (b) Initial discretization of the specimen with 1600 hexahedral elements.

In order to investigate the precision of the adaptive model, numerical modelling with different mesh sizes is implemented. Several internal length scales $l_0 = 0.005 \text{ mm}, 0.01 \text{ mm}$ and the adaptive ratios $c = 2.0, 4.0, 8.0$ are investigated for TAPFM. These analyses are implemented with the fixed adaptive range from $s_{\min} = 0.01$ to $s_{\max} = 0.1$. As a contrast, the example is also modelled with nonadaptive mesh for 2-D and 3-D conditions. For the nonadaptive computations, the internal length scale $l_0 = 0.01 \text{ mm}$ and the minimum element size near the crack $h_{\min} = l_0 / 4$ are applied. The crack paths with adaptive and nonadaptive meshes are shown in **Fig. 7** whereas the load-displacement responses of the specimens are plotted in **Fig. 8-Fig. 9**. It can be seen from **Fig. 7(a)-(b)** and **Fig. 12** that the crack path and force-displacement response of TAPFM are highly consistent with standard PFM so long as the same mesh densities are adopted. The peak force of the loading process for 3-D standard PFM is 15.20 N, while the critical force for TAPFM is 15.21 N, which only results in an error of 0.07%. **Fig. 7(b)-(d)** and **Fig. 9** depict the insensitivity of the model in terms of the internal length scale and mesh size. The peak force tends to converge with the increase of the adaptive ratio c , and the error can be ignored when $c > 4.0$.

Further, the sensitivity analysis in terms of the adaptive range is also carried out with the fixed internal length scale $l_0 = 0.01 \text{ mm}$ and adaptive ratio $c = 4.0$. **Fig. 10** shows the influence of $s_{\max}$ on the force response of the adaptive process when $s_{\min}$ is fixed as 0.01. The force-displacement curve tends to converge by decreasing $s_{\max}$, and a credible result can be obtained if $s_{\max} \leq 0.3$ is adopted. Similarly, the effect of $s_{\min}$ is studied with a fixed $s_{\max} = 0.3$

in **Fig. 11**. The curves become stable as $s_{\min}$ is smaller than 0.05. However, a nonnegligible error can be generated when the overly large $s_{\min}$ or $s_{\max}$ is applied. It is because the adaptation is not conducted in time, and the stress concentration is generated on the crack tip. Hence, the adaptive range $[0.01, 0.1]$ as recommended in Eq. (53) can be adopted to obtain the believable numerical result.

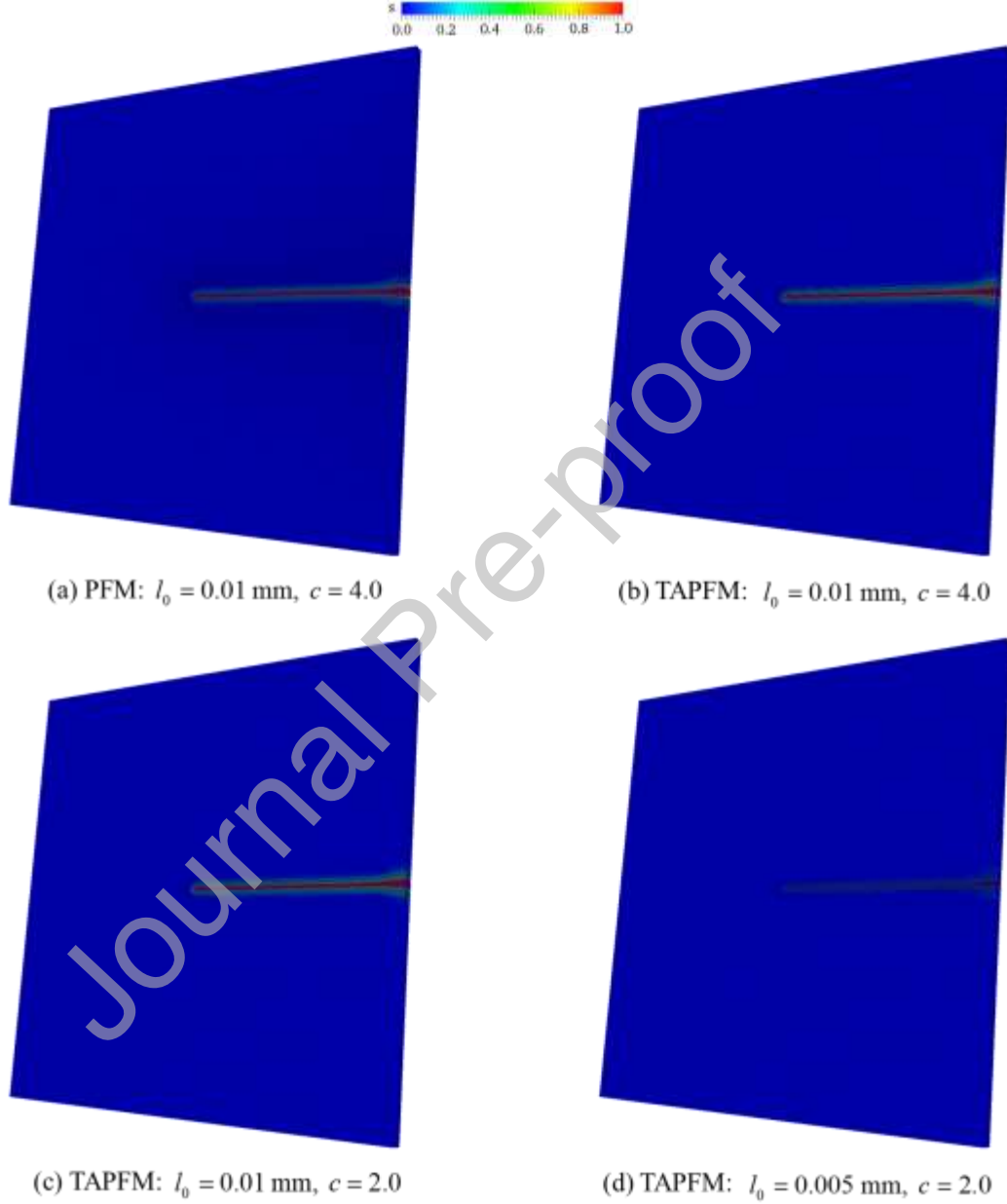


Fig. 7. Single-edge notched plate under mode-I loading: The crack path with (a) pre-refined mesh and (b)-(d) different adaptive mesh ($s_{\min} = 0.01$, $s_{\max} = 0.1$).

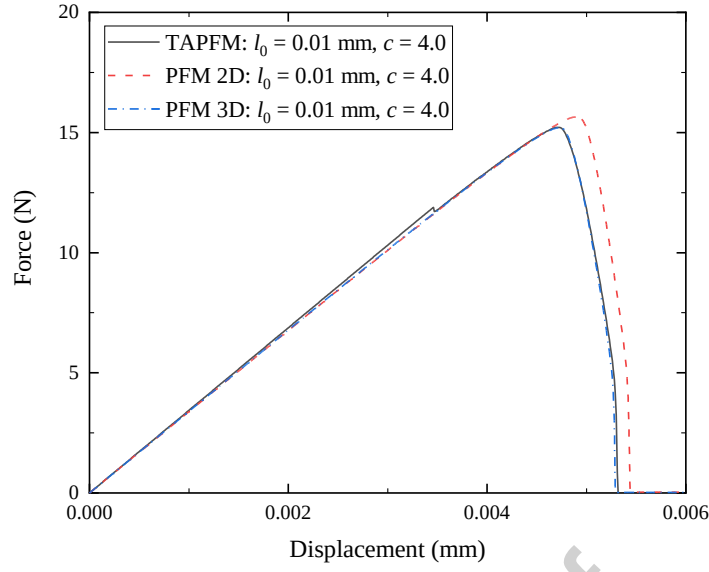


Fig. 8. Single-edge notched plate under mode-I loading: Load-displacement curves for adaptive and nonadaptive modellings ($s_{\min} = 0.01$, $s_{\max} = 0.1$).

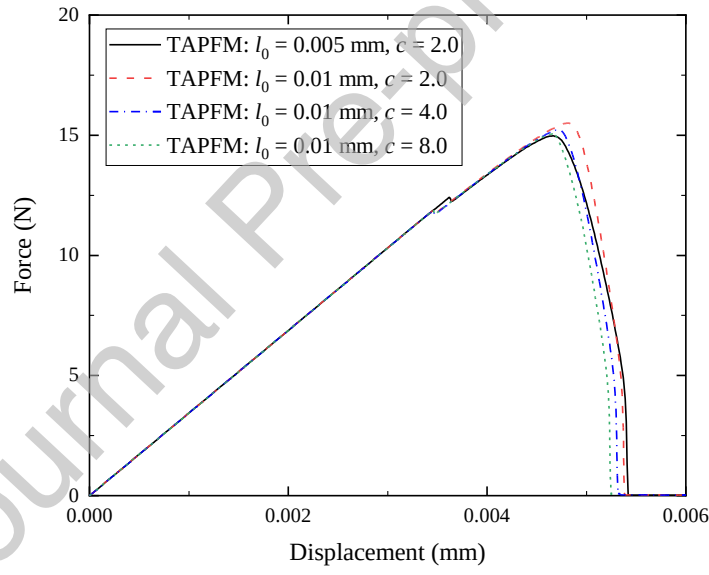


Fig. 9. Single-edge notched plate under mode-I loading: Load-displacement curves with different internal length scales and adaptive ratios ($s_{\min} = 0.01$, $s_{\max} = 0.1$).

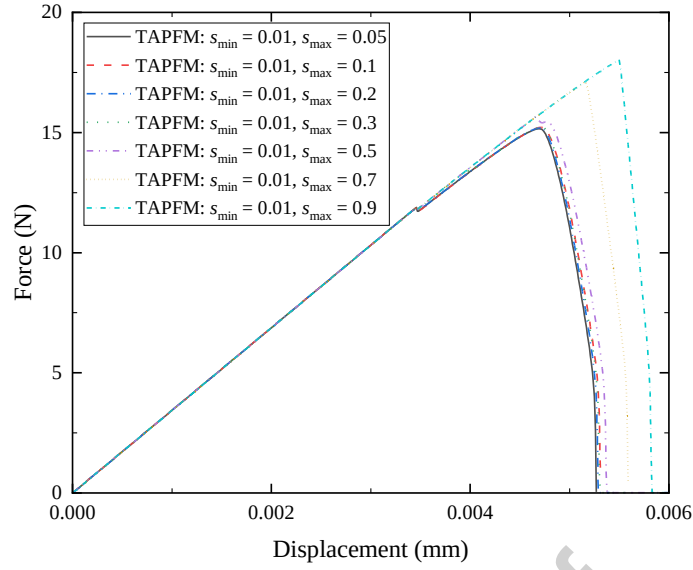


Fig. 10. Single-edge notched plate under mode-I loading: Load-displacement curves with various $s_{\max}$ ($l_0 = 0.01$ mm, $c = 4.0$, $s_{\min} = 0.01$).

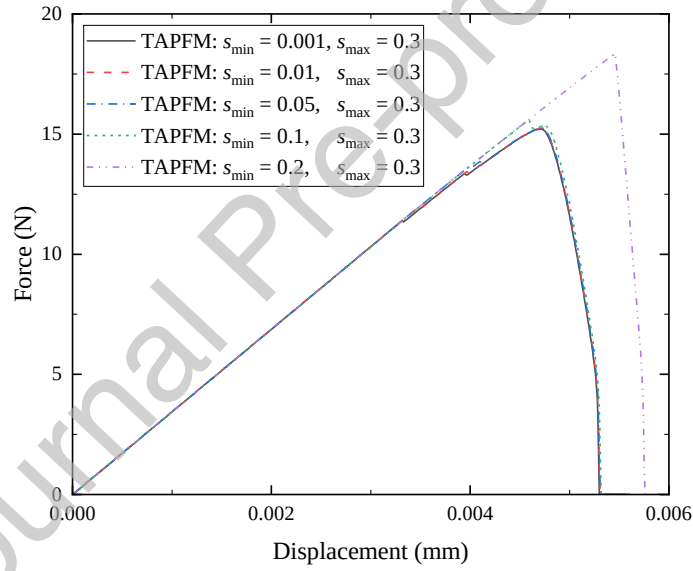


Fig. 11. Single-edge notched plate under mode-I loading: Load-displacement curves with various $s_{\min}$ ($l_0 = 0.01$ mm, $c = 4.0$, $s_{\max} = 0.3$).

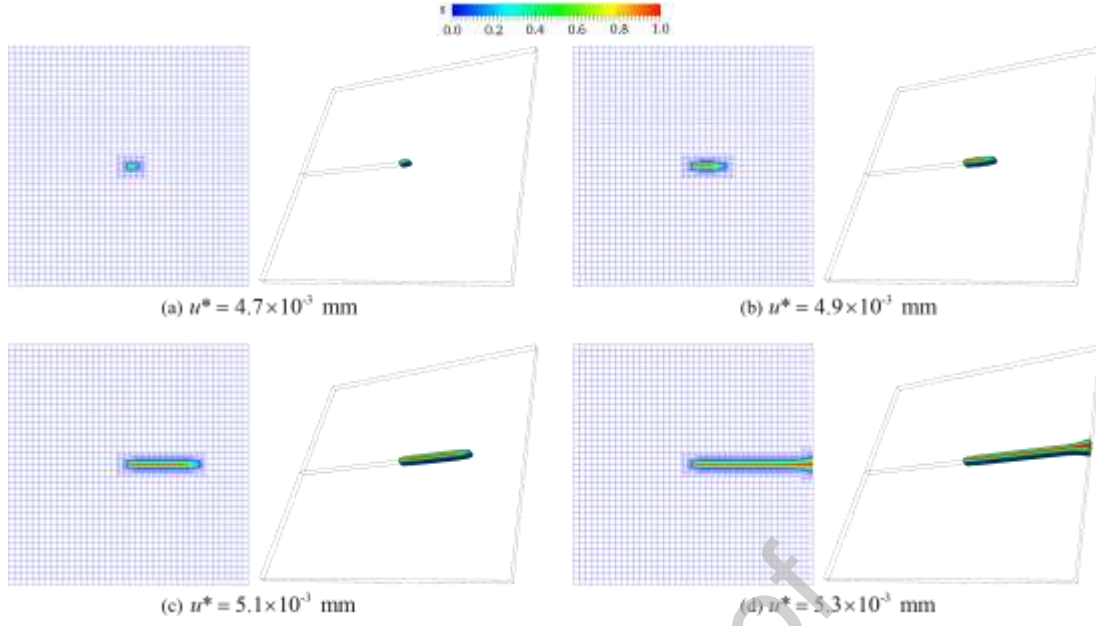


Fig. 12. Single-edge notched plate under mode-I loading: Crack propagation patterns at displacements (a) $u^* = 4.7 \times 10^{-3}$ mm, (b) $u^* = 4.9 \times 10^{-3}$ mm, (c) $u^* = 5.1 \times 10^{-3}$ mm, (d) $u^* = 5.3 \times 10^{-3}$ mm ($l_0 = 0.01$ mm, $c = 4.0$, $s_{\min} = 0.01$, $s_{\max} = 0.1$).

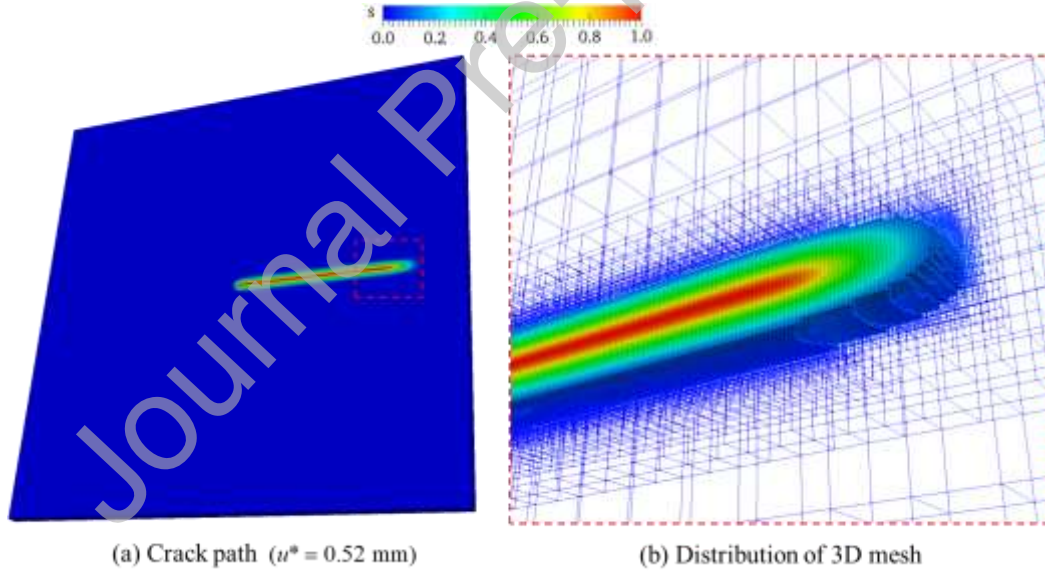


Fig. 13. Single-edge notched plate under mode-I loading: The crack path and three-dimensional mesh distribution on the crack front

$$(l_0 = 0.01 \text{ mm}, c = 4.0, s_{\min} = 0.01, s_{\max} = 0.1, u^* = 5.2 \times 10^{-3} \text{ mm}).$$

Fig. 12 plots various stages of crack propagation and mesh distribution for the parameters $l_0 = 0.01$ mm and $c = 4.0$. It is observed that the mesh refinement only happens in the domain near the crack front. As shown in **Fig. 13**, the smallest elements are only generated in the damaged

area with a limited transition zone. When the plate completely fractures, the final number of elements is 135825 in the adaptive modelling whereas in the case of the standard PFM is 211152. **Fig. 14** compares the computational time for every displacement increment 2×10^{-4} mm and the cumulative time for PFM and TAPFM. It can be seen that the standard PFM needs considerable computational time in every step, while the spent time of each step for TAPFM increases gradually with the evolution of the crack. The average CPU time, required in a step to solve this problem, is 56 min in the standard PFM, whereas is 0.6 min in TAPFM. Hence, substantial memory and CPU time can be saved by using TAPFM.

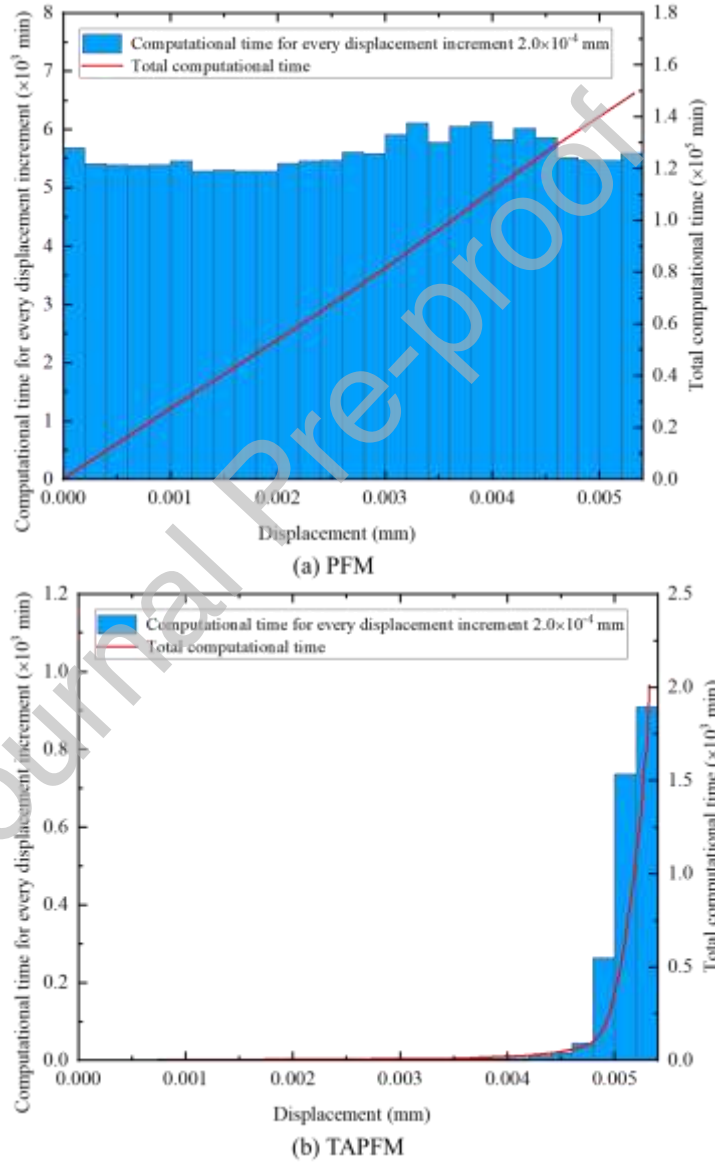


Fig. 14. Single-edge notched plate under mode-I loading: Computational time.

5.1.2. Single-edge notched plate under mode II loading.

This example concentrates on the fracture of the notched plate under mode II loading. A constant rightward displacement increment of $\Delta u_x^* = 5.0 \times 10^{-6}$ mm is imposed on the top face

while the bottom of the specimen is fixed. The vertical displacement of left and right faces is also fixed, as shown in **Fig. 15**. The total displacement is 0.012 mm. The length scale parameters $l_0 = 0.01$ mm and $c = 4.0$ are adopted to ensure the accuracy of the calculation. The mesh discretization is the same as that used in the previous example.

Similarly, the example is simulated by the standard phase-field model for the purpose of comparison, and both 2-D and 3-D nonadaptive simulations are carried out. A nonuniform mesh with the element size $h_{\min} = 0.0025$ mm in the damage domain is adopted in the nonadaptive analyses.

Fig. 16 illustrates the resulting fracture paths for standard phase-field and TAPFM. The crack patterns along with the adaptive mesh at different load steps are shown in **Fig. 17**. The load-displacement behavior of the specimen is depicted in **Fig. 18**. The obtained peak load from 3-D standard PFM is 10.87 N, whereas the critical load of the adaptive modelling is 10.97 N, resulting in an error of 0.92%. As expected, the crack evolution and load-displacement behavior obtained from the TAPFM are in good agreement with the standard phase-field model. Although the 2-D simulation can achieve the similar result to that obtained from 3-D modelling, there is a nonnegligible difference between them in terms of the peak load, which is caused by the deformation in the thickness direction.

To further show the precision of TAPFM, the example is also implemented with a coarser nonadaptive mesh in which $l_0 = 0.02$ mm and $h_{\min} = 0.005$ mm are applied. **Fig. 19** compares the predicted load-displacement curves with different mesh sizes. As can be seen, the same results can be achieved for TAPFM and standard PFM when the identical mesh size is adopted, while the standard PFM with a coarser mesh may lead to a greater error.

Though both adaptive and nonadaptive algorithms reach identical results, their numerical performances are extremely different. For the 3-D standard phase-field model, the plate is discretized into 228200 hexagonal elements with eight nodes. From **Fig. 20**, it can be observed that the total number of elements required in the adaptive method is 172806. Therefore, the system matrix of TAPFM is much smaller compared to standard PFM, which results in very less requirement of computational time and memory space. It is worth noting that, for brittle and quasi-brittle fractures involving fast crack propagation, only a few DOFs are needed for most load steps in TAPFM, which increases rapidly in the final stage of crack propagation. The average computational time in each step required by the standard PFM is 63 min whereas the TAPFM only

needs 4.1 min. In addition, the entire crack can be automatically obtained without any additional preprocessing on mesh in the damage domain, which is a troublesome problem for fracture prediction.

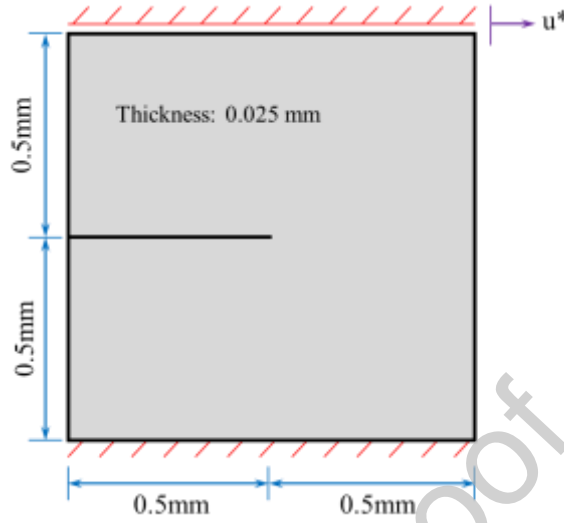


Fig. 15. Single-edge notched plate under mode-II loading: Geometry and boundary conditions.

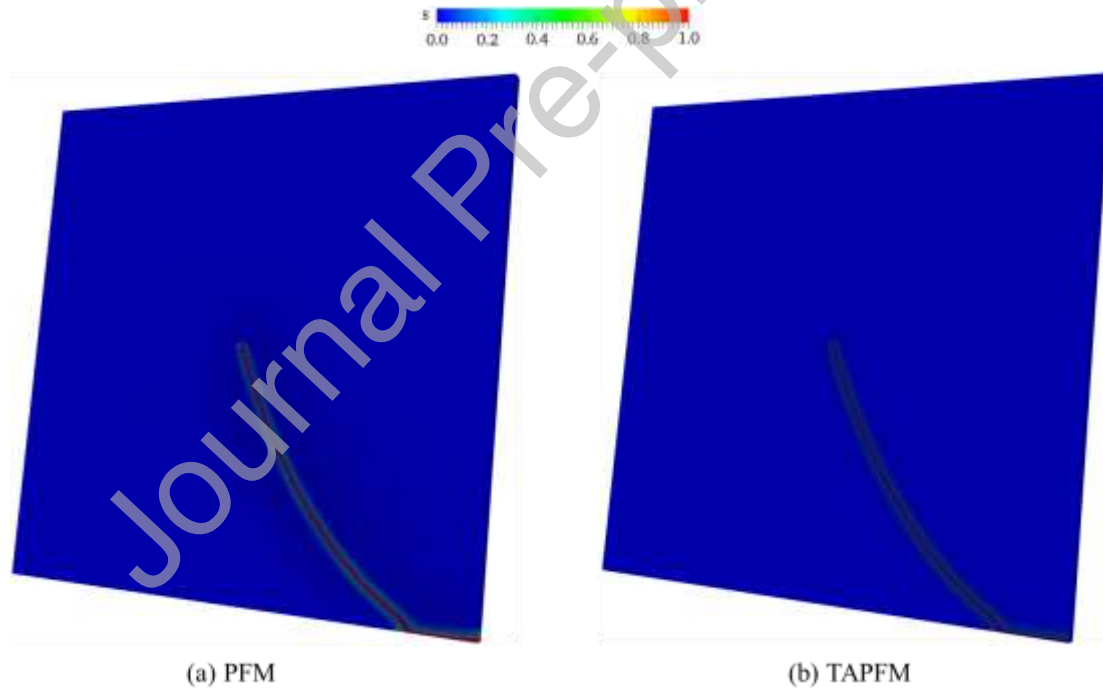


Fig. 16. Single-edge notched plate under mode-II loading: The crack paths for (a) standard PFM and (b) TAPFM.

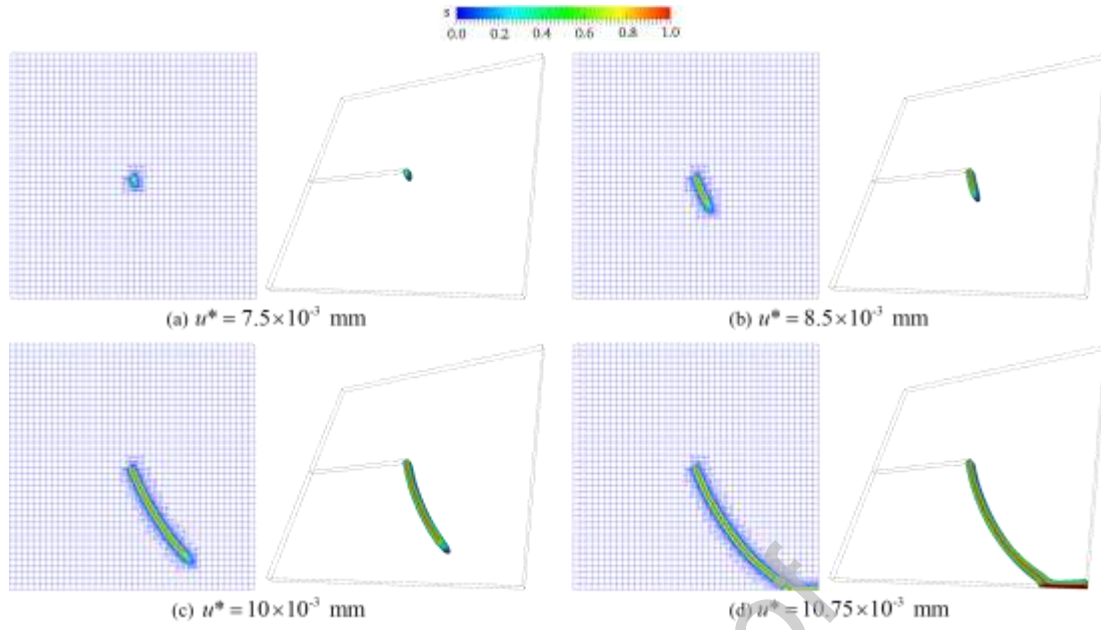


Fig. 17. Single-edge notched plate under mode-II loading: Crack propagation patterns at displacements (a) $u^* = 4.7 \times 10^{-3}$ mm, (b) $u^* = 4.9 \times 10^{-3}$ mm, (c) $u^* = 5.1 \times 10^{-3}$ mm, (d) $u^* = 5.3 \times 10^{-3}$ mm.

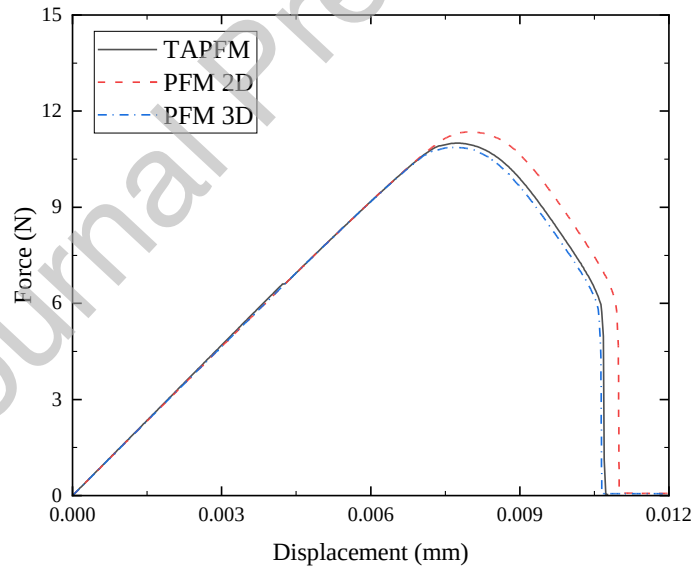


Fig. 18. Single-edge notched plate under mode-II loading: Load-displacement curves with the same mesh size.

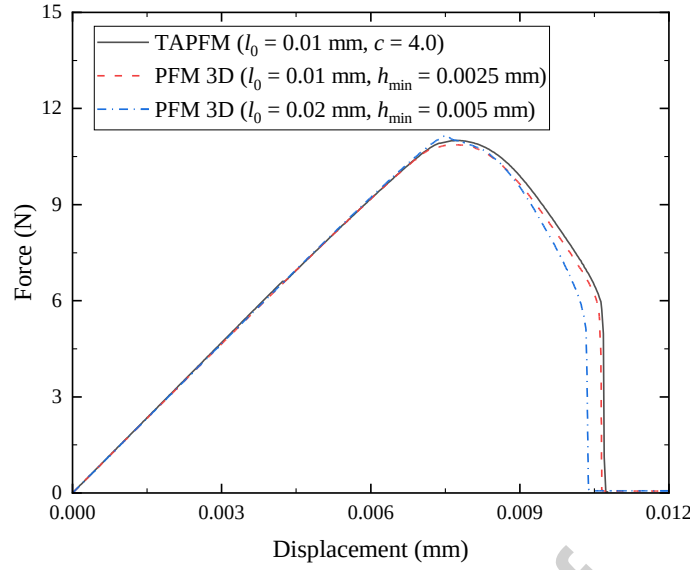


Fig. 19. Single-edge notched plate under mode-II loading: Load-displacement curves with different mesh sizes.

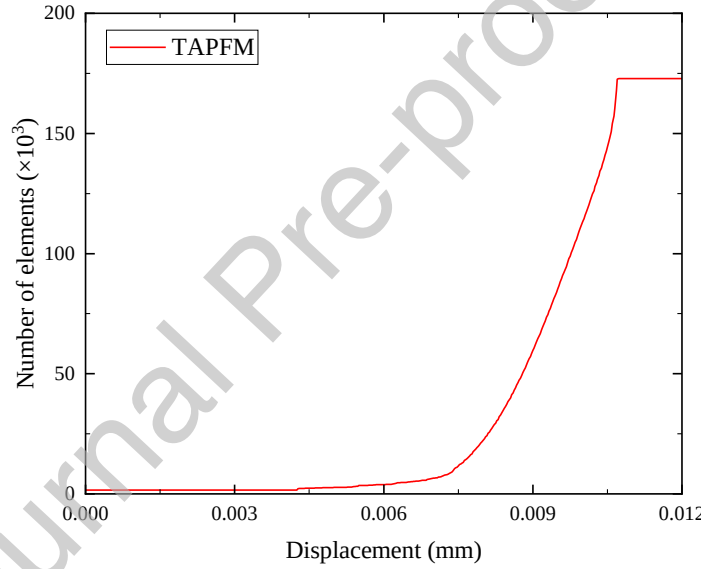


Fig. 20. Single-edge notched plate under mode-II loading: Number of elements.

5.2. Notched beams with different failure modes.

Three notched beams under three-point bending with different material properties and fracture modes are presented in this section. The numerical examples concern mode I, mode I + II, and mode I + III mixed-mode failures, respectively. Moreover, non-adaptive simulations and adaptive analyzes with the same maximum mesh densities are compared. The first two specimens are made of quasi-brittle concrete and the third example adopts the properties of brittle PolyMethyl MethAcrylate (PMMA).

5.2.1. Mode-I fracture of a notched beam under three-point bending.

In this section, a classic three-point bending experiment reported in [62] is modelled first. The

geometry, load, and boundary conditions of the specimen are shown in **Fig. 21**. The thickness of the beam is 20 mm. The notch, which penetrates the bottom of the beam, is 5 mm in width and 50 mm in height.

The beam is made of concrete which is a typical quasi-brittle material, and the Cornelissen's softening curve is assumed. The material parameters are taken from [62]: Young's modulus $E_0 = 20$ GPa , Poisson's ratio $\nu_0 = 0.2$, failure strength $f_t = 2.4 \times 10^{-3}$ GPa , and fracture toughness $G_c = 1.13 \times 10^{-4}$ kN/mm . Besides, the length scale $l_0 = 1$ mm is assumed. A prescribed downward displacement $u^* = 1.0$ mm is applied with the uniform incremental displacement $\Delta u^* = 5 \times 10^{-4}$ mm . As shown in **Fig. 22**, the domain is discretized by 31552 coarse-scale elements with the element size $h = 1$ mm near the notch for TAPFM in the initial stage. However, the discretization for the standard PFM requires 175856 elements.

Fig. 23 and **Fig. 24** present the damage profiles obtained by the standard PFM and the TAPFM, respectively. Finally, the number of elements reaches 63997 accompanied by adaptive mesh refinement, as shown in **Fig. 24(b)**. It takes 42.1 min and 2.5 min for the standard PFM and TAPFM, respectively, in terms of the average time in a step. The load-displacement curves for standard PFM and TAPFM are plotted in **Fig. 25**. As can be seen, the difference between nonadaptive and adaptive simulation results is also small. The error of the peak load between standard PFM and TAPFM is only 0.9%.

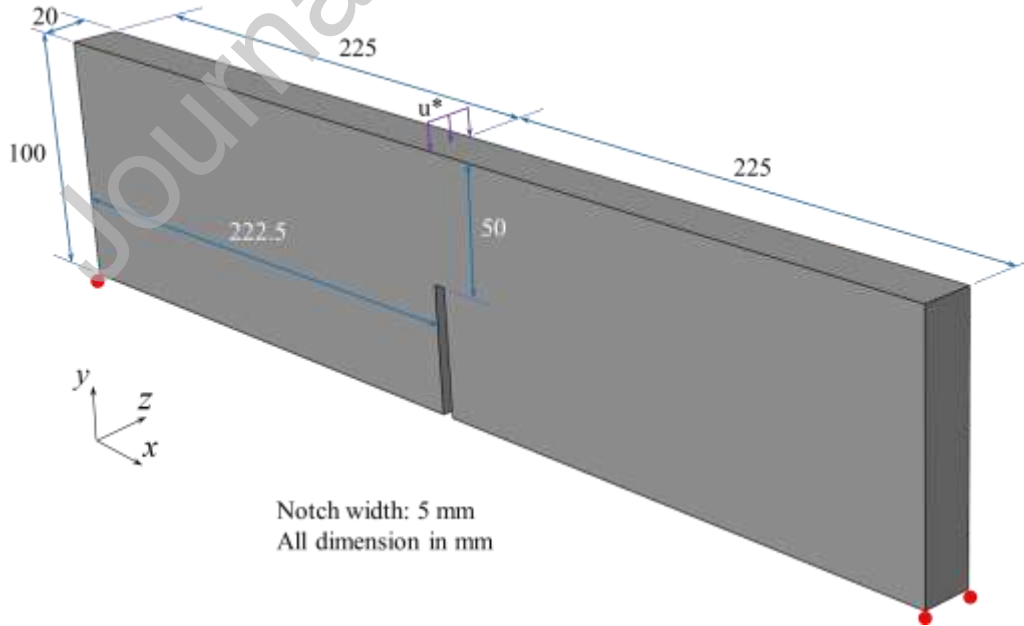


Fig. 21. Notched beam under three-point bending: Geometry and boundary conditions.

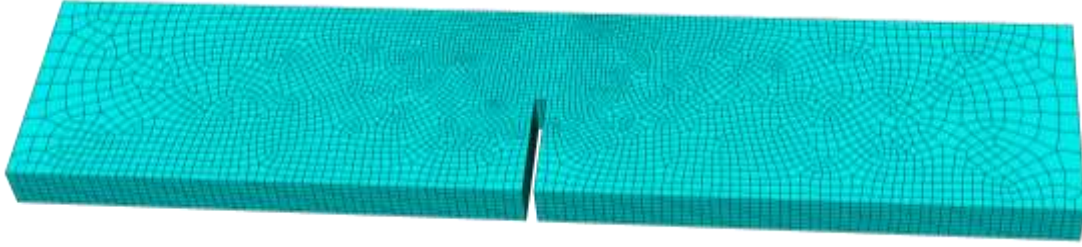


Fig. 22. Notched beam under three-point bending: 3-D initial finite element discretization (31552 eight-node hexahedral elements).

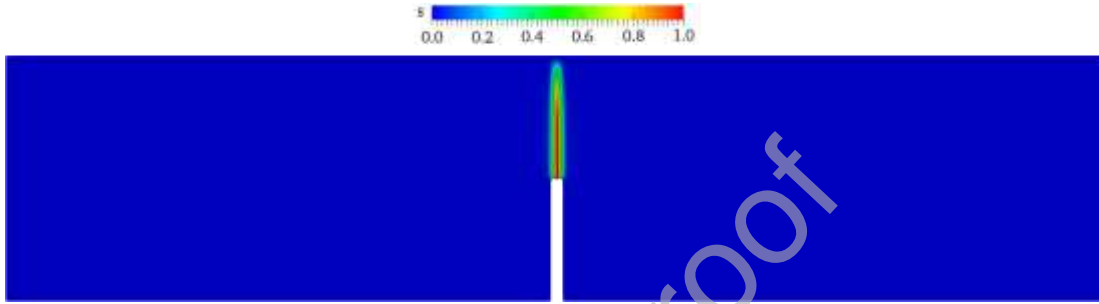


Fig. 23. Notched beam under three-point bending: Crack pattern using the standard FEM.

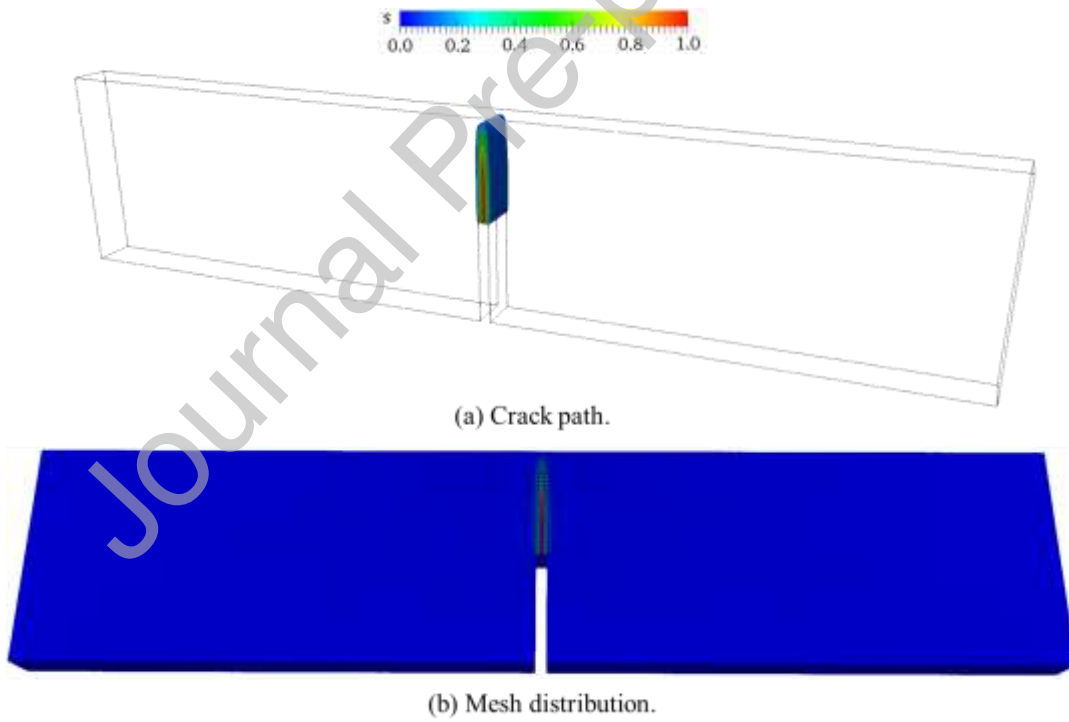


Fig. 24. Notched beam under three-point bending: The (a) crack pattern and (b) mesh refinement at $u^* = 1$ mm using TAPFM.

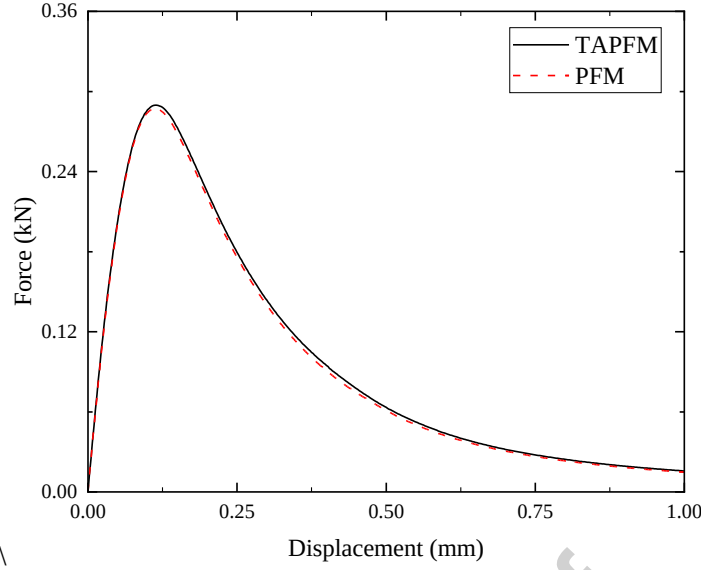


Fig. 25. Notched beam under three-point bending: Load-displacement curves.

5.2.2. Mode-I + II fracture of a notched beam under eccentric bending.

This example is to further demonstrate the capability of the proposed TAPFM in simulating mode-I + II damage. This basic numerical problem was originally introduced by [63] and consequently has been implemented by [64] using the peridynamic model. The schematic of the specimen with the initial defect subjected to loading and constraints is depicted in **Fig. 26**. As can be seen, the beam has a total length of 675 mm, an effectively supported span of 375 mm, a total height of 150 mm, and a uniform thickness of 20 mm. In **Fig. 27**, the initial mesh, consisting of 19852 coarse-scale eight-node hexagonal elements, with the size $h=5$ mm near the notch, is presented. The specimen is also implemented using the standard PFM, with a mesh size of $h_{\min} = 0.5$ mm in the damage domain.

The beam is also made of concrete so that the Cornelissen softening curve is applied in the numerical simulation. As in [35], the following material properties are adopted: Young's modulus $E_0 = 38$ GPa, Poisson's ratio $\nu_0 = 0.2$, tensile strength $f_t = 3 \times 10^{-3}$ GPa, and fracture toughness $G_c = 6.9 \times 10^{-5}$ kN/mm. The length scale is considered as $l_0 = 2.5$ mm to obtain a clear crack path. A downward displacement $u^* = 0.15$ mm is imposed on the top face of the beam with a constant increment $\Delta u^* = 2.5 \times 10^{-4}$ mm. During the loading process, the crack mouth opening displacement (CMOD) is monitored.

The crack patterns obtained from the standard PFM and the proposed TAPFM are presented in **Fig. 28** and **Fig. 29**, respectively. **Fig. 30** compares the load-CMOD curves using the standard PFM and TAPFM. The error caused by the adaptation is 0.38% in terms of the peak load. Similar to the previous examples, the adaptive strategy gives identical results to the nonadaptive

simulation with the same mesh density.

In the numerical modelling, the required number of elements corresponding to the maximum refined area is 398240 for the standard PFM and 176953 for the TAPFM. Moreover, the average computational time of PFM and TAPFM for one step is 121 min and 15.2 min, respectively. Obviously, much time and memory can be saved in the whole calculation.

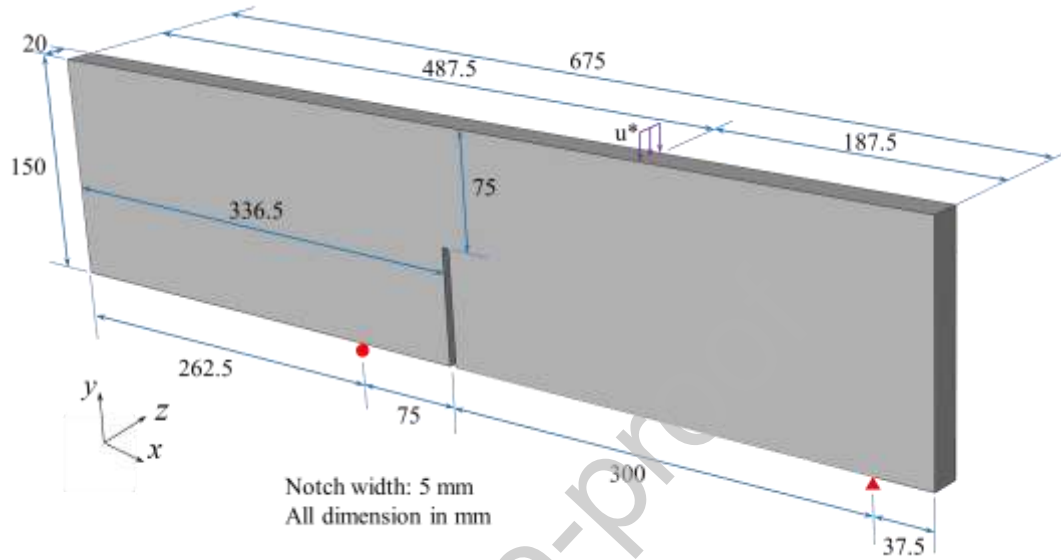


Fig. 26. Notched beam under eccentric bending: Geometry and boundary conditions.

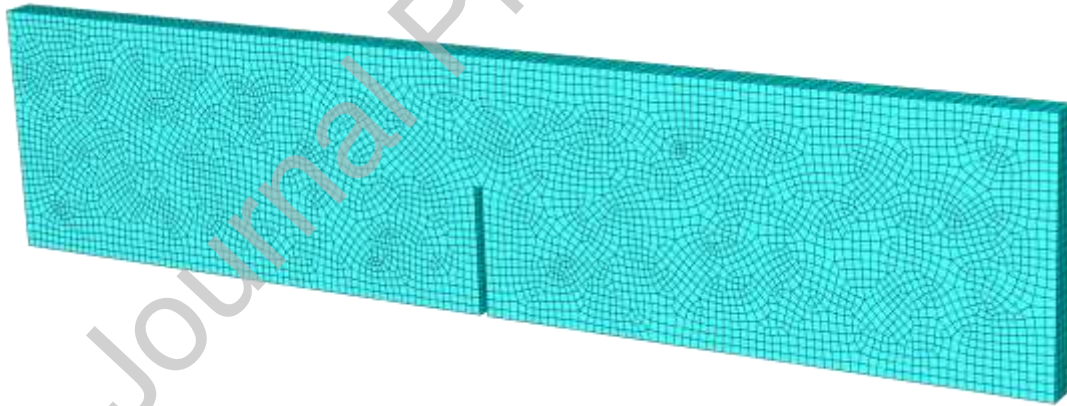


Fig. 27. Notched beam under eccentric bending: 3-D finite element discretization (19852 eight-node hexahedral elements).

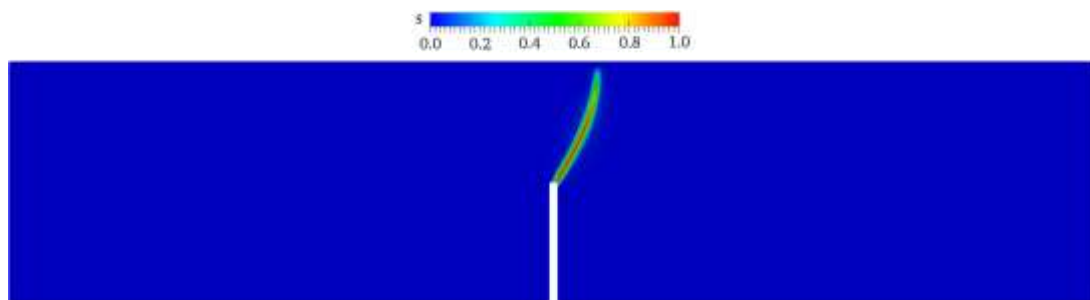


Fig. 28. Notched beam under eccentric bending: Crack pattern in the standard FEM.

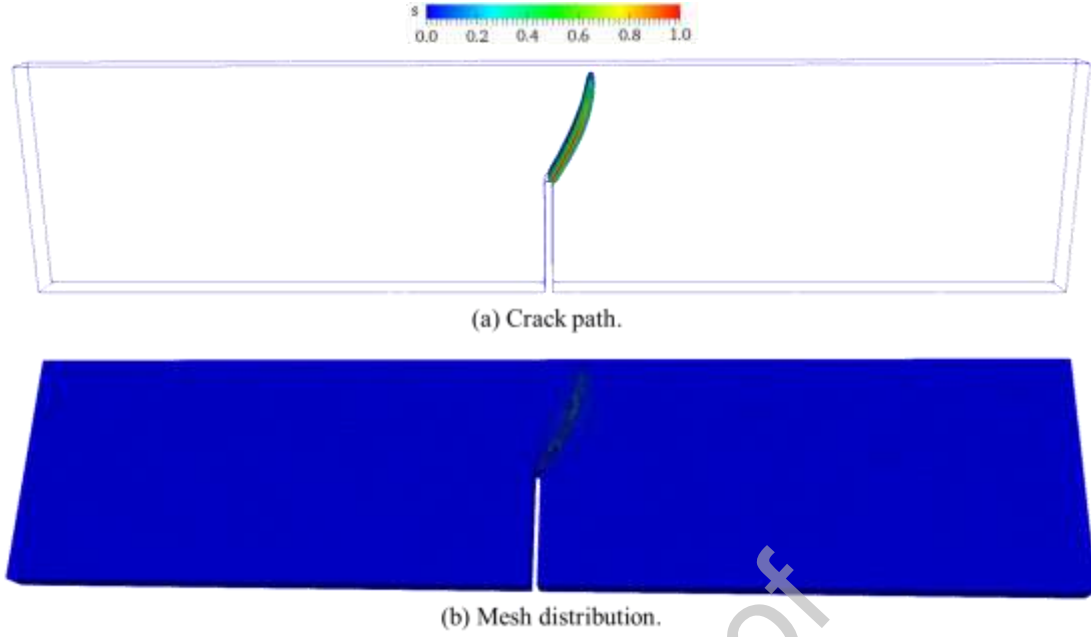


Fig. 29. Notched beam under eccentric bending: The (a) crack path and (b) mesh distribution at

$$u^* = 0.15 \text{ mm.}$$

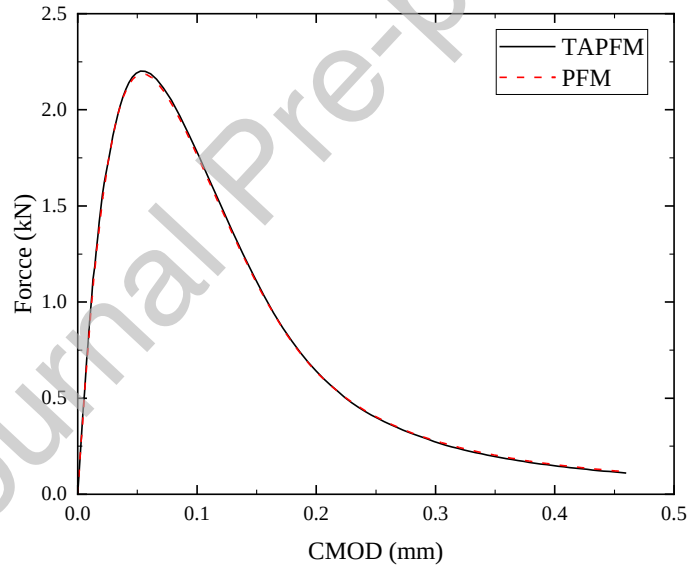


Fig. 30. Notched beam under eccentric bending: Load-CMOD curves.

5.2.3. Mode-I + III fracture of a skew notched beam under three-point bending.

Here, the experimental study [65, 66] of the mixed-mode crack in a skew notched beam is used as a numerical benchmark. As shown in **Fig. 31**, the beam is of dimensions $260 \times 60 \times 10 \text{ mm}^3$ with an effectively supported span of 240 mm. A zero-thickness flaw with a height of 20 mm and an inclination angle of 45° is located at the bottom of the specimen.

The specimen is made of PMMA, which is known as a typical brittle material, so that the linear softening curve is considered. The material parameters are taken from [67, 68]: Young's

modulus $E_0 = 2.8 \text{ GPa}$, Poisson's ratio $\nu_0 = 0.38$, tensile strength $f_t = 0.02 \text{ GPa}$, and fracture toughness $5 \times 10^{-4} \text{ kN/mm}$. The internal length scale is taken as $l_0 = 1 \text{ mm}$. Besides, the displacement control strategy is applied to obtain the load-displacement behavior. A constant displacement increment $\Delta u^* = 5 \times 10^{-4} \text{ mm}$ is used up to $u^* = 0.75 \text{ mm}$.

Fig. 32 and **Fig. 33** illustrate the initial and final mesh distributions of the beam, respectively. The beam is originally discretized by 4428 8-node hexahedral elements with their size $h \approx 5 \text{ mm}$. The number of elements keeps constant until the crack appears, and then grows rapidly to 201870 for the complete damage state. Compared with the final mesh, a tremendous number of elements can be saved in most load steps.

The crack paths obtained from the experimental [65] and numerical results are compared in **Fig. 34** and **Fig. 35**. It can be observed that the crack starts with a spatial twist from the 45° slanted notch to the symmetric plane, and then propagates vertically along the direction of loading. The transition from mixed-mode to mode-I failure is well captured by the proposed model. In **Fig. 36**, the relative positions of the crack tip on the front surface are compared with test data [67]. Obviously, the computed crack pattern is in good accordance with that from the experiment.

The perspective views of the damage profile at various heights of the specimen are presented in **Fig. 37**. As mentioned before, the crack rotates from the top of the notch until it recovers mode-I fracture, and then continues to develop upward in the mid-plane.

It should be noted that the crack front is not a straight but a curve line in most cases, as shown in **Fig. 35**. The left and top views of the crack front at various loading stages are plotted in **Fig. 38** (a) and **Fig. 38** (b), respectively. In the side view, the shape of the crack front varies from a concavity to a plane, and then to a convexity. As for the vertical view, a distinct anti-symmetric crack kinking along the inclined notch can be observed, with the kink angles $0 \leq \theta \leq 45^\circ$ for $z > 0$ and $-45 \leq \theta \leq 0^\circ$ for $z < 0$. As expected, the qualitative evolution of the crack front for the experiment is accurately reproduced by the TAPFM.

Finally, the load-displacement behavior of the skew notched beam obtained by the adaptive method is depicted in **Fig. 39**. The brittle characteristic of PMMA is well exhibited in the curve. The force response remains an almost linear elastic development until it reaches the peak load. Then, a cliff-like drop of the force appears due to the rapid initiation and propagation of the crack. Hence, the experimental results are perfectly reflected once again.

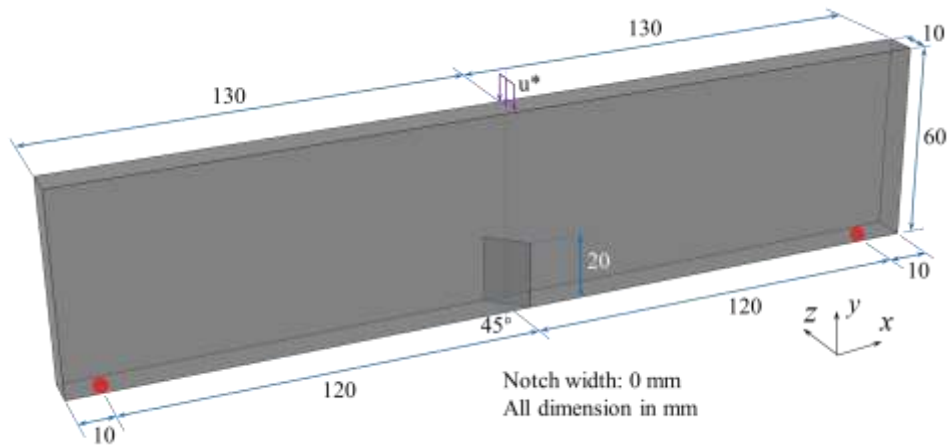


Fig. 31. Skew notched beam under three-point bending: Geometry and boundary conditions.

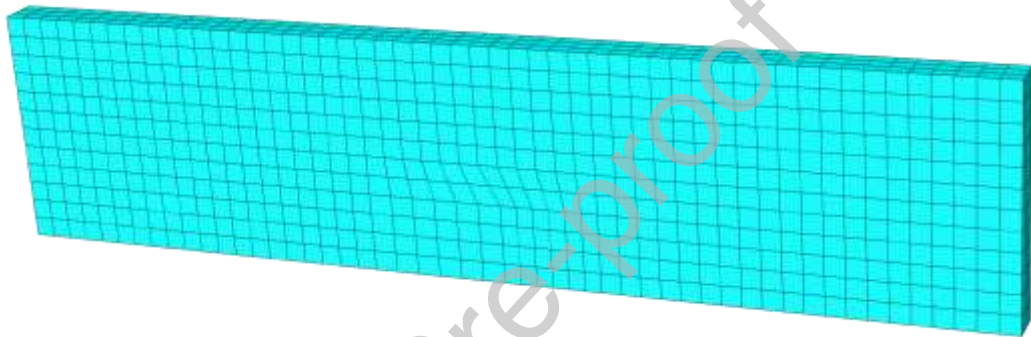


Fig. 32. Skew notched beam under three-point bending: 3-D finite element discretization (4428 eight-node hexahedral elements).

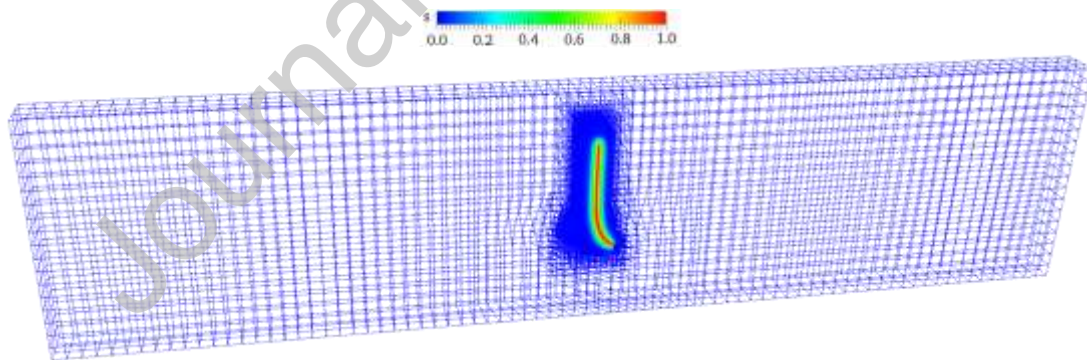


Fig. 33. Skew notched beam under three-point bending: Mesh refinement.

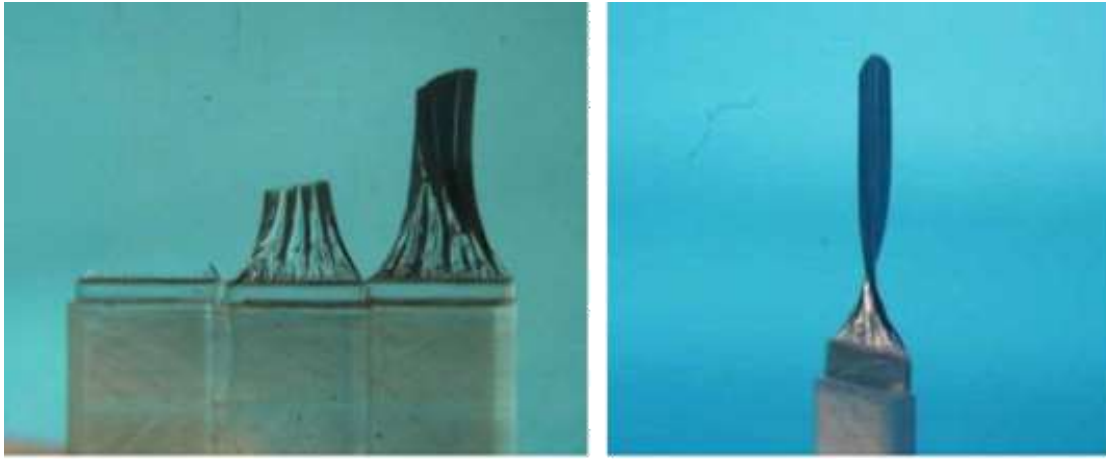


Fig. 34. Skew notched beam under three-point bending: Experimental results [65].

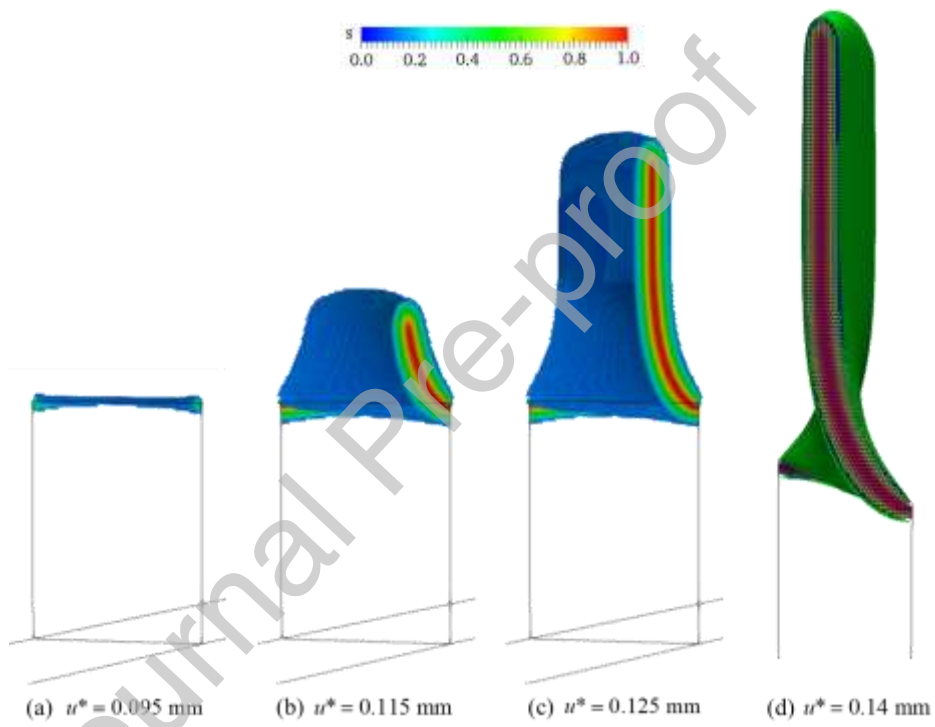


Fig. 35. Skew notched beam under three-point bending: Crack propagation of numerical simulation.

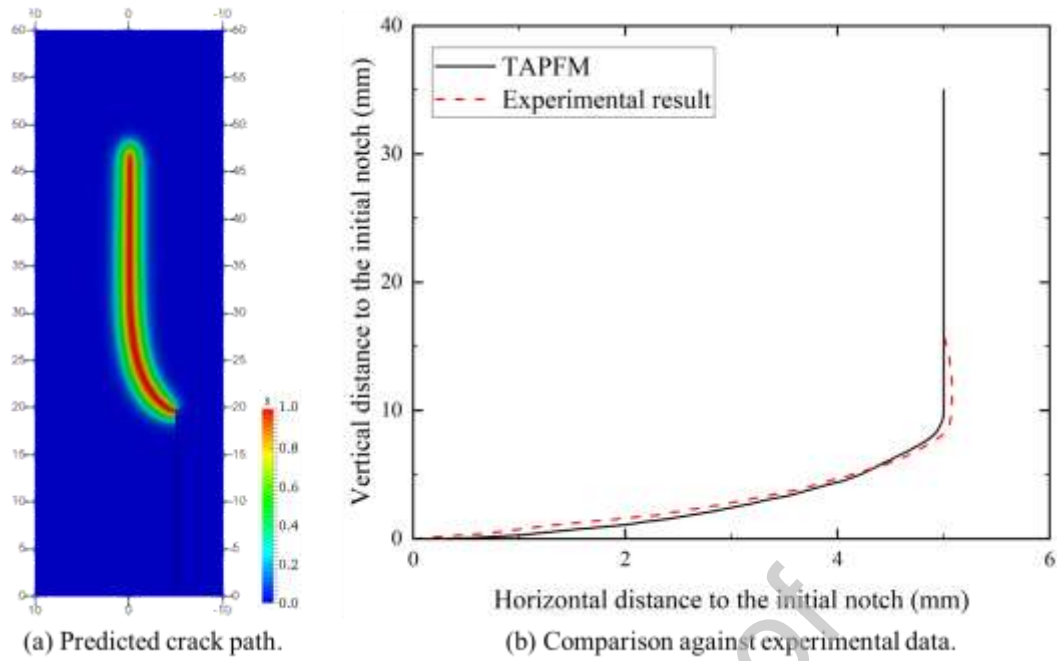


Fig. 36. Skew notched beam under three-point bending: Crack pattern on the front surface.

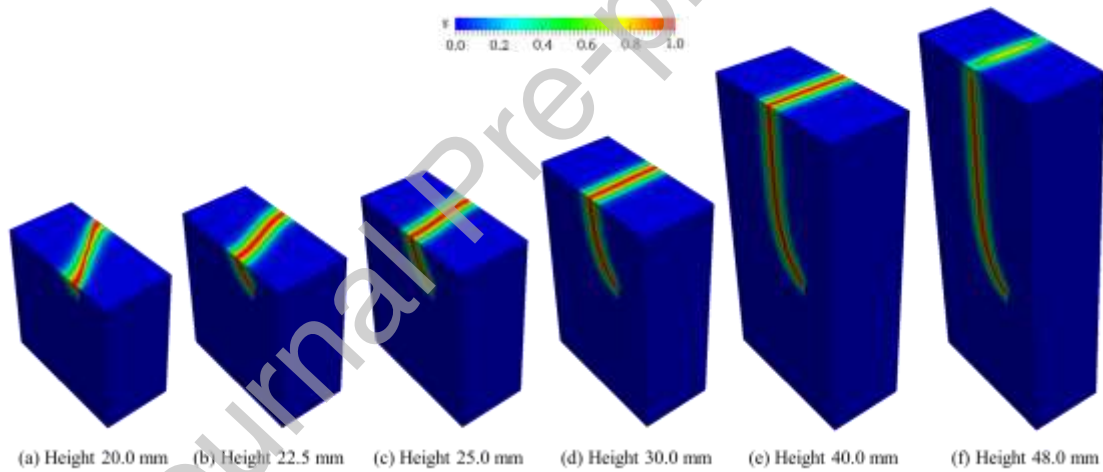


Fig. 37. Skew notched beam under three-point bending: Predicted crack pattern at various heights of the specimen (perspective view).

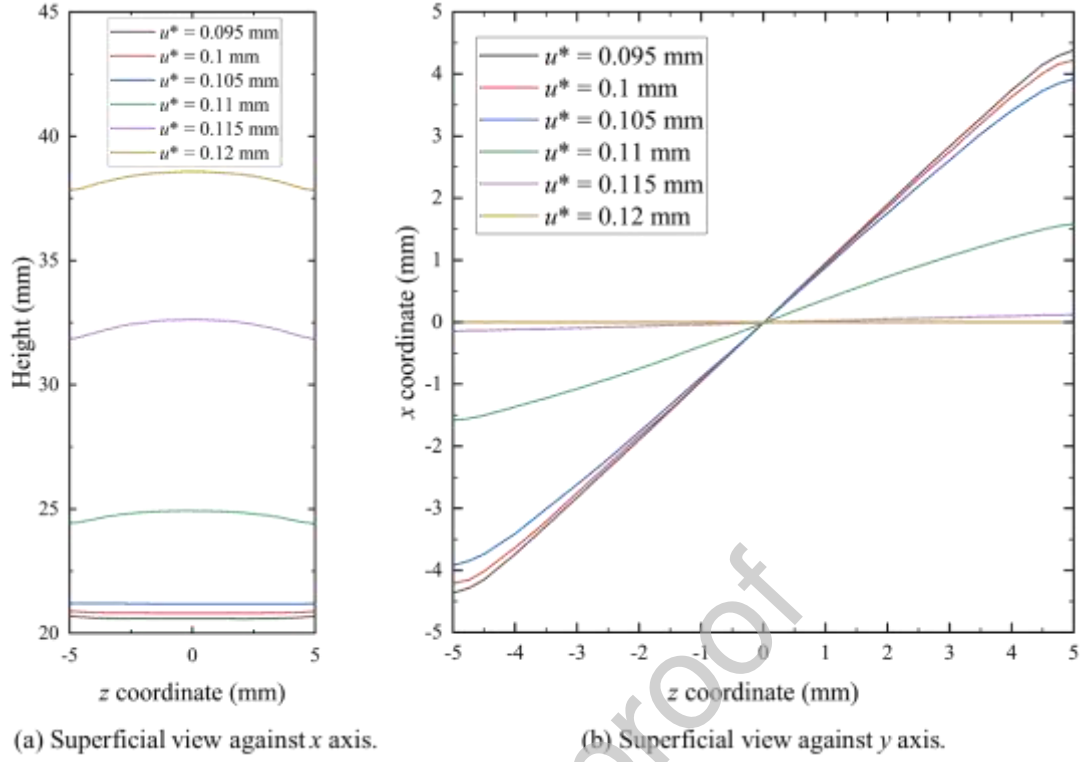


Fig. 38. Skew notched beam under three-point bending: Profiles of crack fronts.

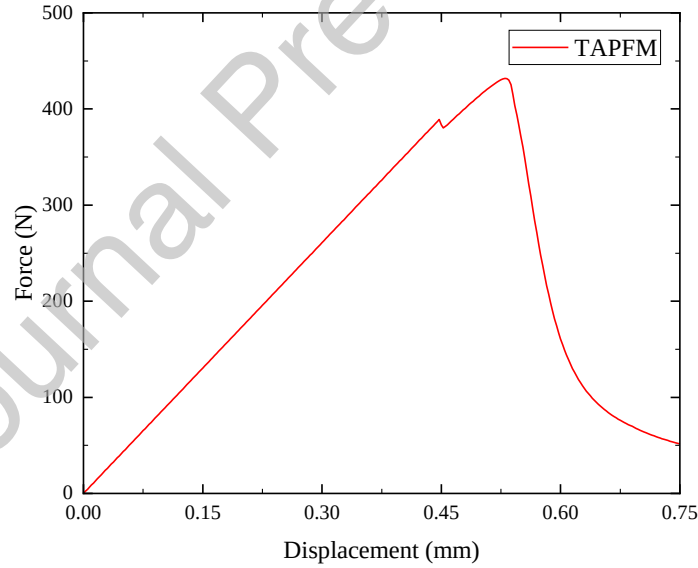


Fig. 39. Skew notched beam under three-point bending: Force-displacement curve.

5.3. Prismatic skew notched beam under torsion.

Here, to further investigate the performance of the proposed method for 3-D fracture problems, the example concentrates on a prismatic skew notched beam submitted to torsion. The experiments were first carried out by [69], and the corresponding numerical reproductions were conducted by [67, 70]. From **Fig. 40**, details of the geometry, loading, and boundary conditions can be extracted. In the experiment, the rectangular beam of dimensions $450 \times 100 \times 100 \text{ mm}^3$

is positioned horizontally, with an inclined notch on its top half. The steel frames clamped at both ends of the specimen are assumed as rigid bodies in the numerical implementation. A downward displacement $u^* = 0.75 \text{ mm}$ is applied on one arm of a steel frame with the constant increment $\Delta u^* = 1 \times 10^{-3} \text{ mm}$, whereas the other three steel arms are restrained in the vertical direction.

The adopted material properties are [71]: Young's modulus $E_0 = 35 \text{ GPa}$, Poisson's ratio $\nu_0 = 0.2$, tensile uniaxial strength $f_t = 2.3 \times 10^{-3} \text{ GPa}$, and fracture toughness $G_c = 8 \times 10^{-5} \text{ kN/mm}$. The internal length scale is applied as $l_0 = 3 \text{ mm}$. Considering the experiment was mostly performed on plain concrete specimens in most cases, the Cornelissen softening curve can be assumed for this example. The whole structure is divided into 42400 initial hexahedral elements with an average size of $h = 5 \text{ mm}$, as shown in **Fig. 41**.

The mesh distribution corresponding to the final failure, with the maximum number of elements taken as 294498, is presented in **Fig. 42**. The crack evolution is always accompanied by adaptive progress, as depicted in **Fig. 43**, to ensure that the crack propagates in the right direction. Nevertheless, only quite a small number of elements in the failure domain are refined without losing accuracy.

The final crack path of the numerical simulation is illustrated in **Fig. 44**. As can be seen, a skew-symmetric crack starts from the notch almost vertically, and then runs through the whole specimen. The computed crack surface is compared with that obtained from the experimental tests on PMMA, as shown in **Fig. 45**. A global S-shape of the damage profile, with a spatially twisted and warped crack surface, can be observed in both numerical and experimental results [72]. In **Fig. 46**, the experimental crack surfaces [69] of the concrete specimen are further compared with the numerically computed ones. As expected, the complex twisting pattern of the crack is perfectly reproduced.

Finally, **Fig. 47** shows the force versus CMOD plots. The numerical solution is compared with the experimental data [68] on the whole loading history. As expected, the prediction of force-CMOD curve is in good accordance with experimental results, though there is difference of peak load which may result from the error of measured tensile strength. The ability and accuracy of the TAPFM in modelling 3-D fracture are perfectly verified once again.

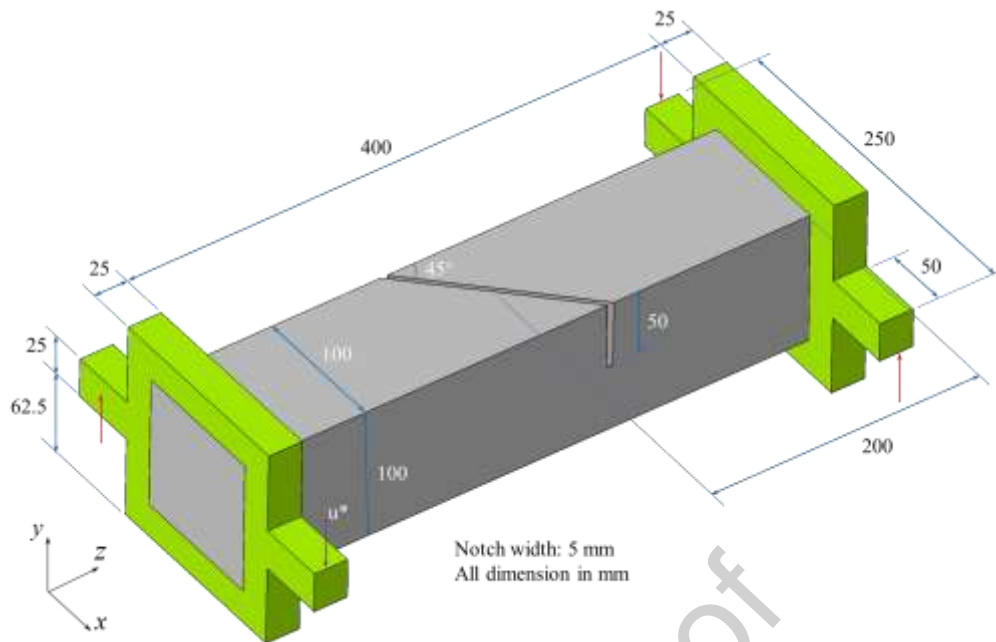


Fig. 40. Prismatic skew notched beam under torsion: Geometry and boundary conditions.

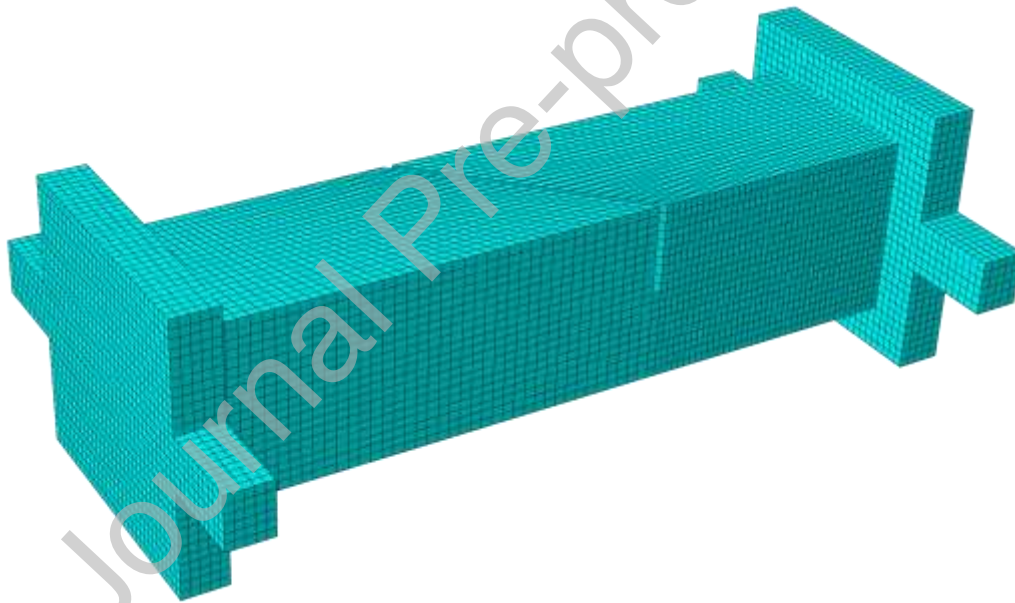


Fig. 41. Prismatic skew notched beam under torsion: 3-D finite element discretization (42400 eight-node hexahedral elements).

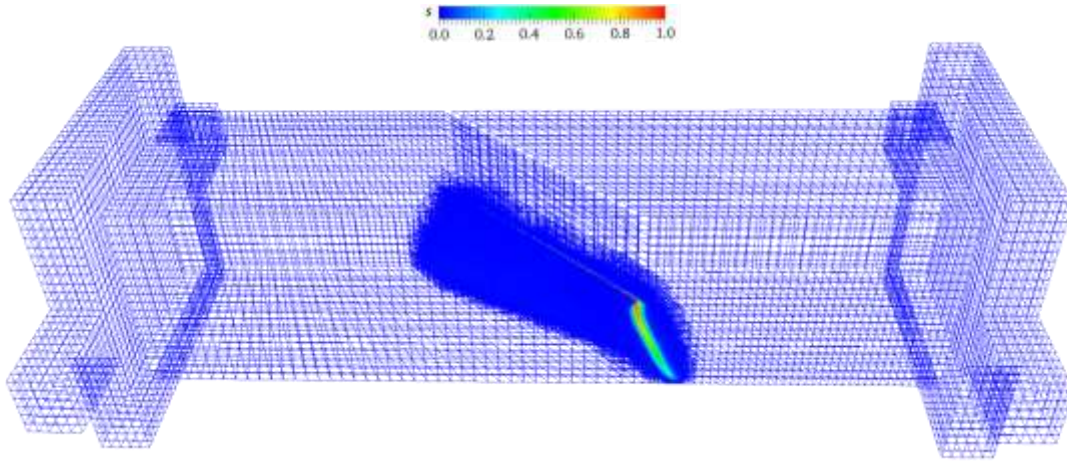


Fig. 42. Prismatic skew notched beam under torsion: Mesh refinement.

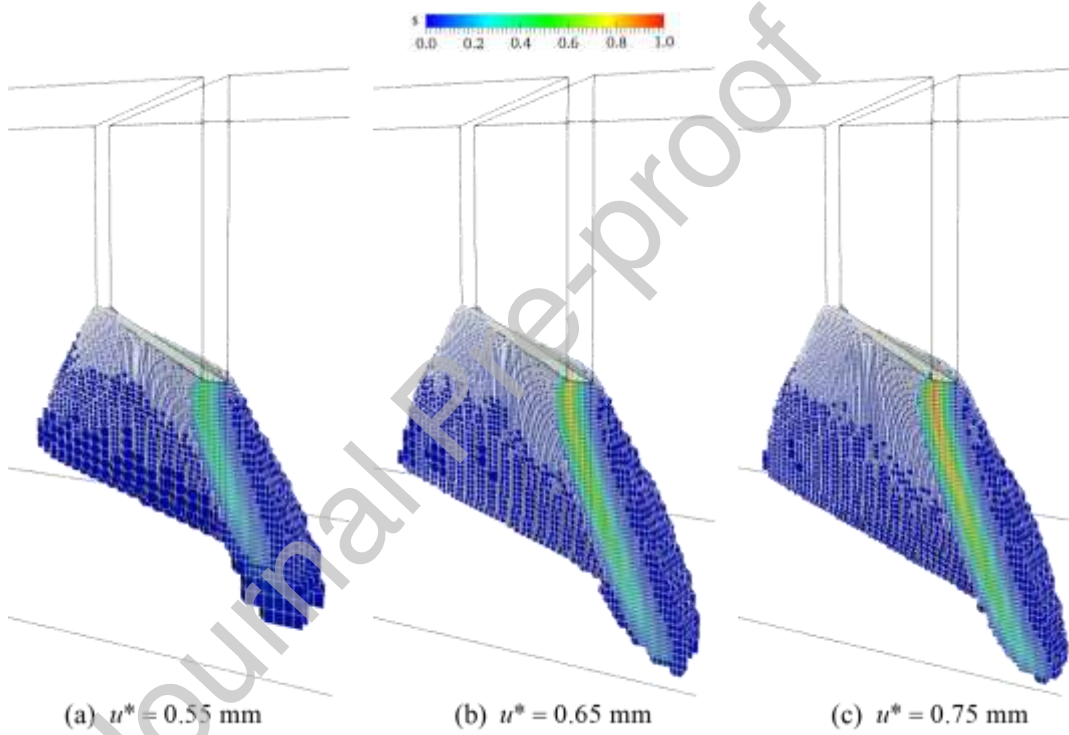


Fig. 43. Prismatic skew notched beam under torsion: Crack propagation with mesh adaptation.

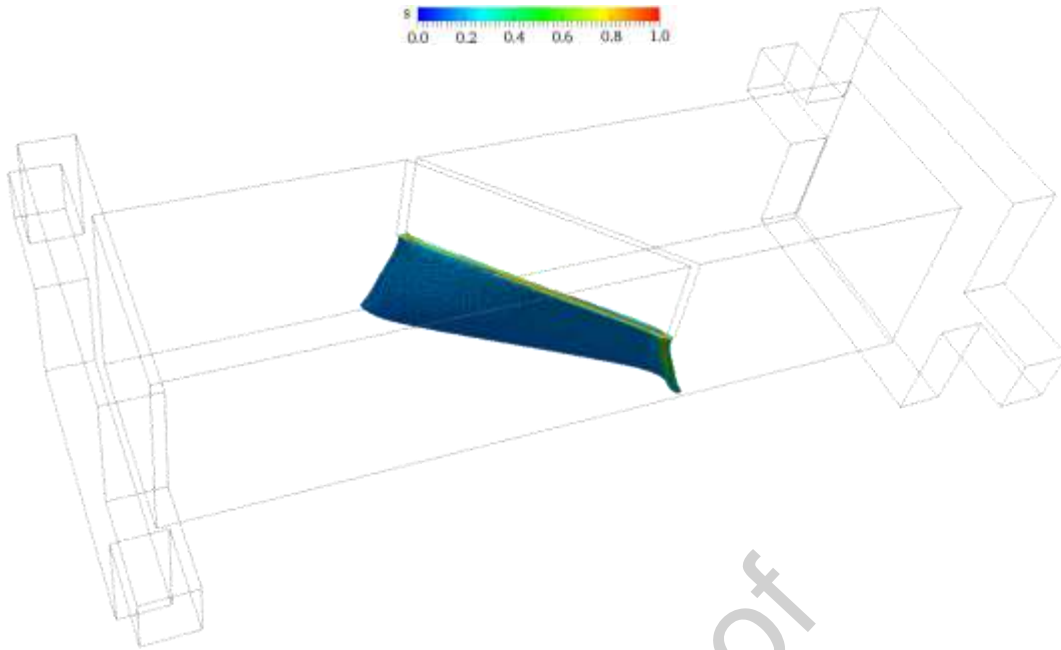
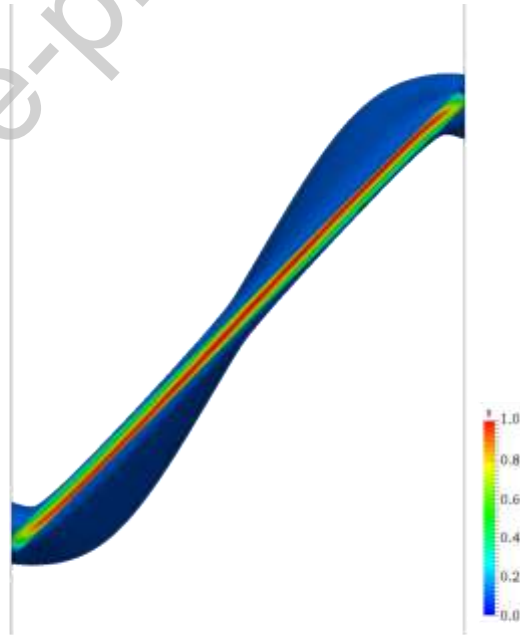


Fig. 44. Prismatic skew notched beam under torsion: Perspective view of crack path.



(a) Experimental crack of PMMA.



(b) Numerical crack.

Fig. 45. Prismatic skew notched beam under torsion: Twisted crack surface.

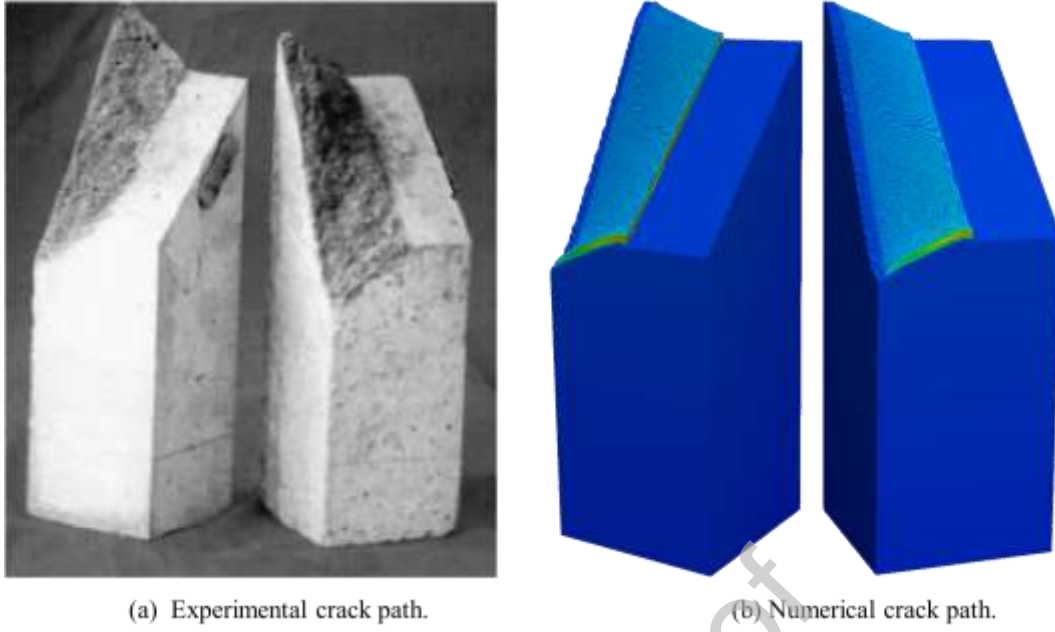


Fig. 46. Prismatic skew notched beam under torsion: Fractured specimen.

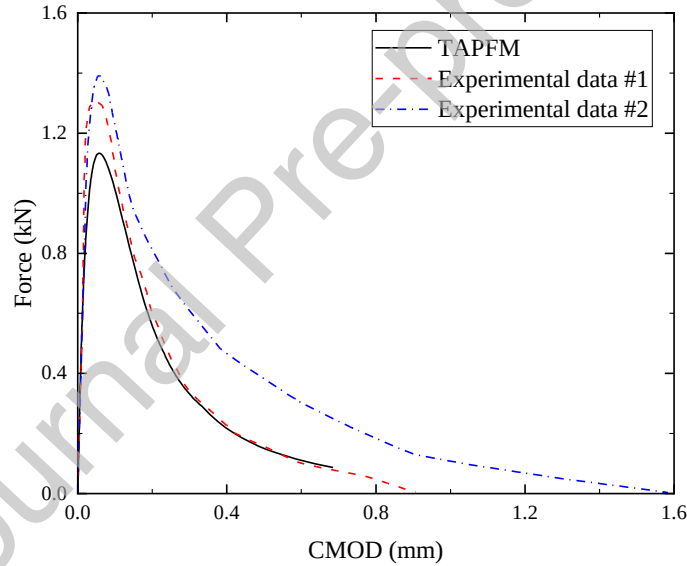


Fig. 47. Prismatic skew notched beam under torsion: Load-CMOD curves of numerical and experimental tests [68].

5.4. Prismatic specimen with two internal flaws.

The goal of this example is to study the branching and merging behaviors of three-dimensional fracture. As shown in **Fig. 48**, it is a prismatic specimen of dimensions $70 \text{ mm} \times 70 \text{ mm} \times 20 \text{ mm}$, with two initial cracks of the same sizes $30 \text{ mm} \times 10 \text{ mm}$ inside the specimen. The left and right flaws are perpendicular and parallel to the bottom surface of the cuboid, respectively. The bottom boundary of the model is vertically constrained, while the left surface is fixed in the horizontal direction. Besides, an upward displacement $u^* = 0.08 \text{ mm}$ is

imposed monotonically in 800 equal increments on the top surface, and a rightward displacement with the same value is applied on the right surface.

The mechanical properties of the material are taken as: Young's modulus $E_0 = 30 \text{ GPa}$, Poisson's ratio $\nu_0 = 0.2$, tensile uniaxial strength $f_t = 3.0 \times 10^{-3} \text{ GPa}$, and fracture toughness $G_c = 1.1 \times 10^{-4} \text{ kN/mm}$. The internal length scale is considered as $l_0 = 2 \text{ mm}$, and the linear softening behavior is assumed in the simulation. As illustrated in **Fig. 48** (b), the specimen is initially discretized by hexahedral elements with uniform mesh size $h = 2 \text{ mm}$.

Fig. 49 presents the final computational mesh obtained by the proposed model. As expected, the mesh refinement is only conducted in the damage subdomain. The crack profiles for different loading stages are plotted in **Fig. 50**. It can be seen that the damage first emerges at the boundary of the initial flaws. As the imposed displacement increases, an arc-shaped crack generates from the axial plane of the specimen, which gradually connects the two flaws together. Then, the branching behavior can be observed at the junctions between the bridge crack and the initial flaws. The branch that originated from the left flaw propagates horizontally, while another branch on the right flaw extends along the vertical direction. With the further rise of the load, the damage continuously evolves and meanwhile, cracks propagate in the thickness direction, tending to penetrate the entire specimen. When the boundary displacement arrives at $u^* = 0.08 \text{ mm}$, the elastic body is separated into four parts by the final cracks. The numerical curve of force versus applied displacement is plotted in **Fig. 51**. At the damage stage, the coalescence and bifurcation of the cracks result in the bends of the curve.

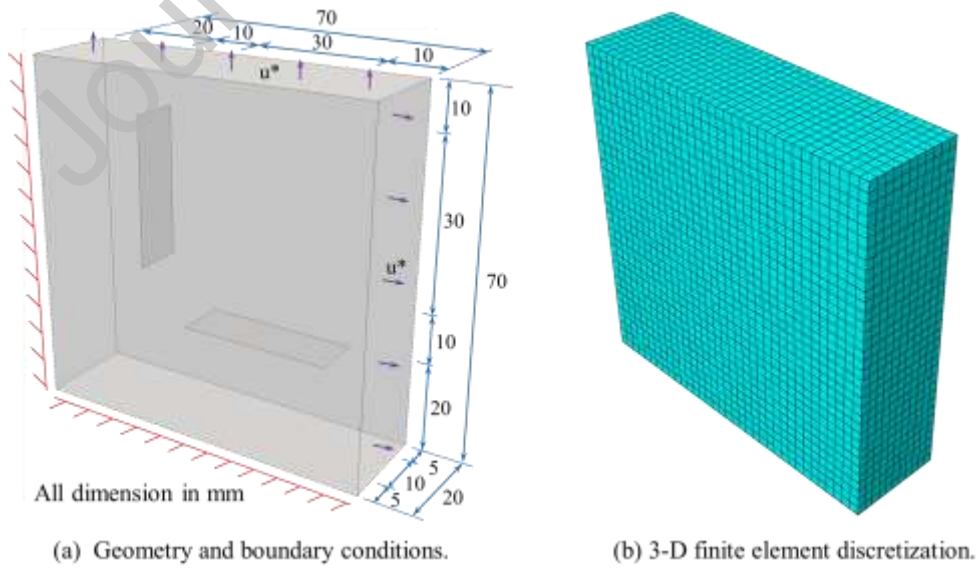


Fig. 48. Prismatic specimen with two internal flaws: Geometry, boundary conditions, and finite element mesh.

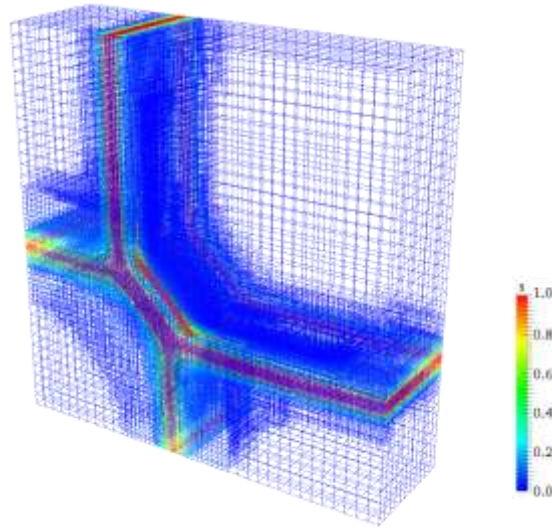


Fig. 49. Prismatic specimen with two internal flaws: Mesh refinement.

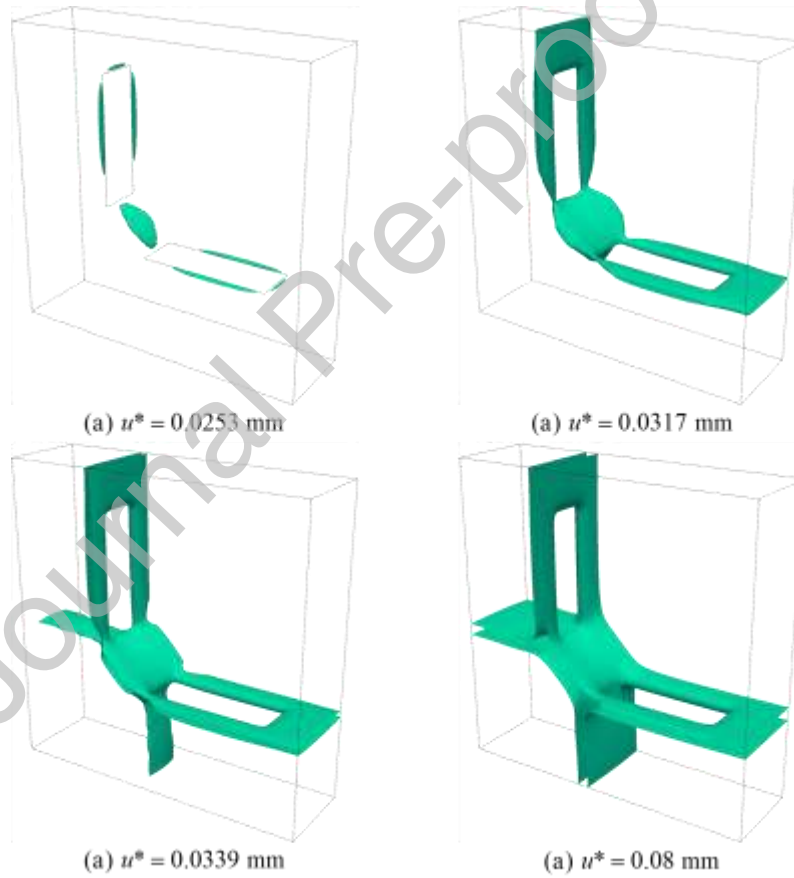


Fig. 50. Prismatic specimen with two internal flaws: Crack propagation of numerical simulation.

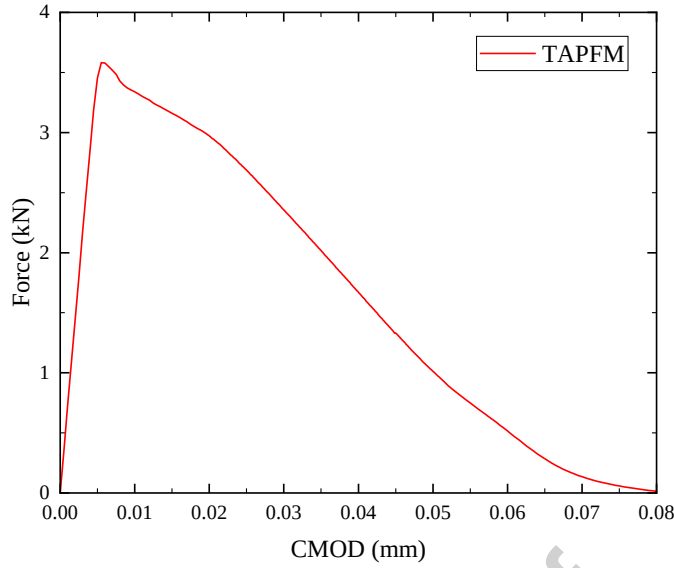


Fig. 51. Prismatic specimen with two internal flaws: Force-displacement curve.

6. Conclusions

In this paper, a novel adaptive method for phase-field model, called the trilinear element-based adaptive phase-field model (TAPFM), is proposed to model the 3-D fracture problems efficiently. In TAPFM, the adaptive strategy based on trilinear multi-node elements is coupled with the length scale insensitive phase-field model in finite element framework to achieve optimal mesh refinement. Crack initiation and propagation can be successfully simulated by the outlined method. In particular, the method shows attractive efficiency and accuracy in modelling brittle and quasi-brittle fracture. According to the representative numerical experiments, the main findings can be summarized as follows:

(1) By using the trilinear shape functions, an arbitrary number of elements can be inserted into the element without increasing its order. The proposed adaptive method requires much fewer elements than the standard phase-field model, which leads to an acceptable cost in terms of memory and CPU time.

(2) The proposed adaptive criterion establishes a connection between the ultimate mesh size and the length scale in the damage domain. It guarantees that a proper mesh density can be obtained once the length scale is determined, although the initial mesh before adaptation may be nonuniform.

(3) The TAPFM is insensitive to the internal length scale, which is a troublesome parameter for classical phase-field models, due to the combined length scale insensitive phase-field model. Hence, the stable crack path and mechanical responses can be obtained for various length scales.

Moreover, this characteristic also loosens the restriction for mesh size determination in numerical modelling.

(4) The comparison with the standard phase-field model as well as experiments demonstrates the accuracy of the proposed method. The presented model not only shows its excellent capability of reducing computational costs, but also guarantees attractive precision in terms of force response. It is proved that the numerical results are in good accordance with that obtained from experimental tests.

(5) The adaptive technique is competent in reproducing the complex 3-D crack path for several fracture patterns, including Mode I, I+II, and I+III fracture. All the major features of three-dimensional fracture can be captured by the method, such as the twist of crack surface and the evolution of crack front.

In addition to these advantages, the refinement strategy keeps tracking the crack tip automatically and only performs the refinement in the vicinity of the crack. That is to say, the prediction of the damage domain for spatial discretization is no longer prior required in the modelling of fracture. In general, the proposed TAPFM would be very useful in analyzing those problems which require large computational effort.

Although the creditable result can be obtained by the proposed method, a slight error can be produced due to the staggered solving algorithm. A more accurate model based on iterative strategy is under development.

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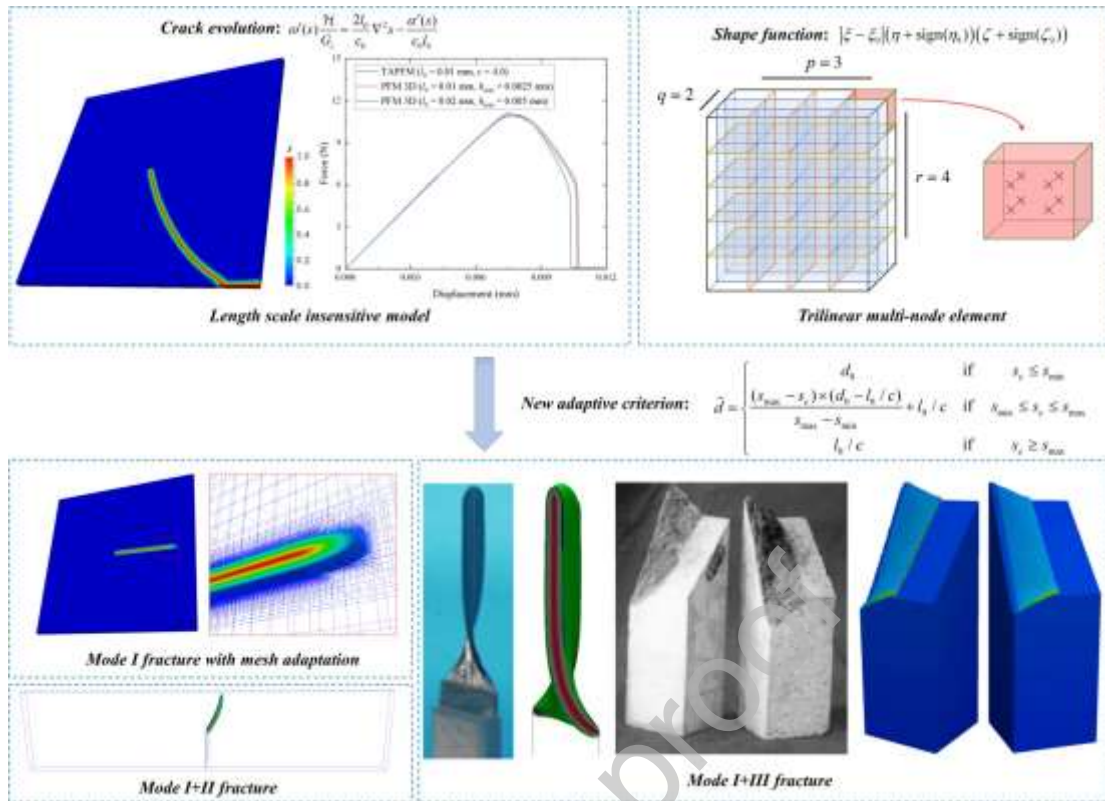
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Graphical Abstract



Credit author statement

Qiang Yue: Methodology, Software, Writing – Original Draft, Formal analysis, Visualization.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this

paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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